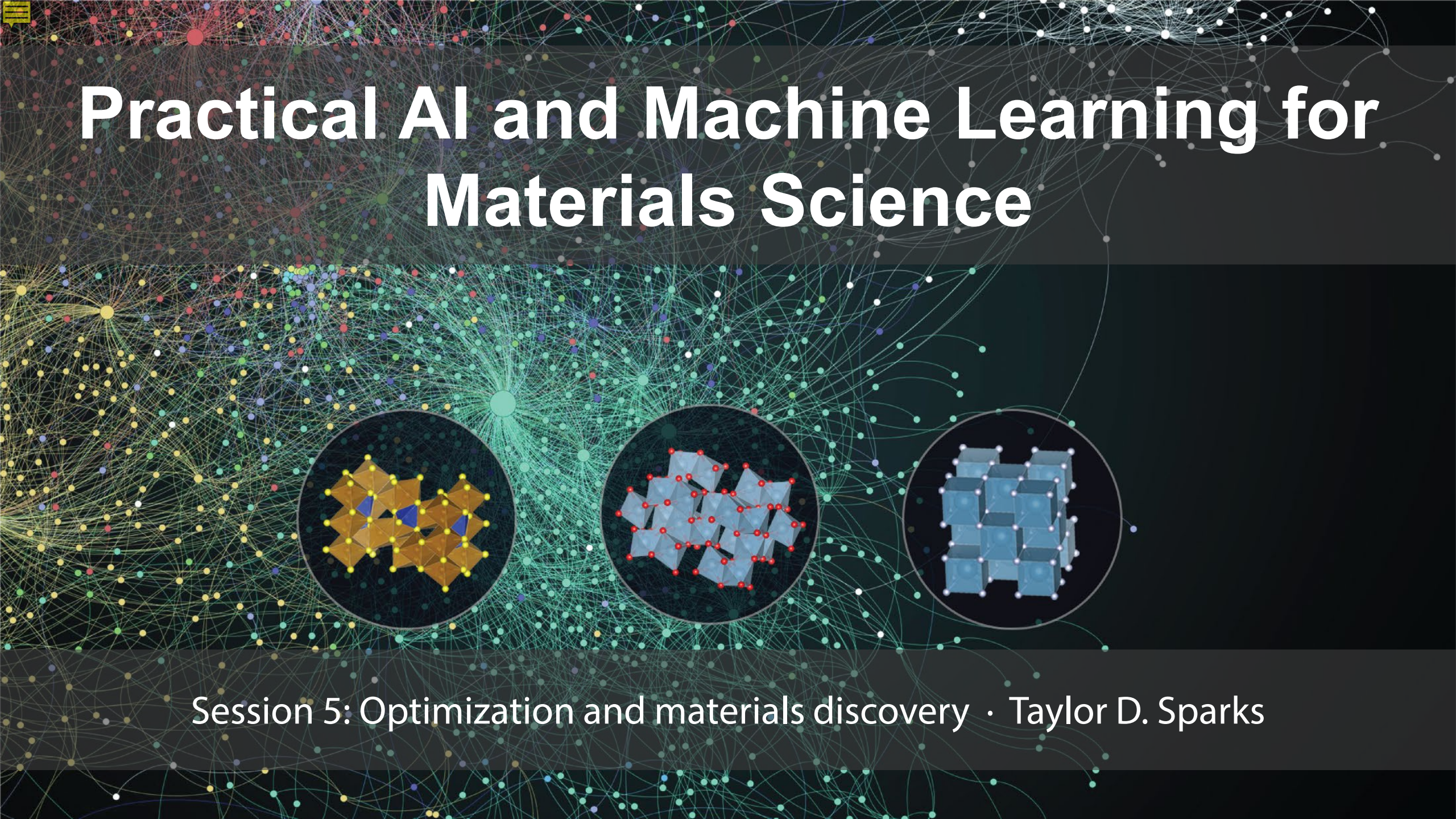
Practical AI and Machine Learning for Materials Science



Session 5: Optimization and materials discovery · Taylor D. Sparks

Last session: four things to remember

1. Check the physics

Redundant quantities catch errors statistics can't

2. Importance is a hypothesis

It describes the model, not nature

3. Physics helps outside the data

Inside it, it mostly buys interpretability

4. Validate measurements too

Image-derived features set the model's ceiling

Today: what should I make next?

Optimizing a process or recipe was your most common goal

Process settings
Sintering T and time, growth conditions, print speed

Formulations
Glazes, coatings, admixtures: fractions add to 100%

Batches
A furnace load or a spin-coating session at a time

Several goals
Performance and cost, supply risk, stability

New materials
Which thermoelectric to make next?

Simulations
CFD, DFT and MD as cheap 'experiments'

One loop handles all of them: build a model → suggest an experiment → update

Why optimize with a model?

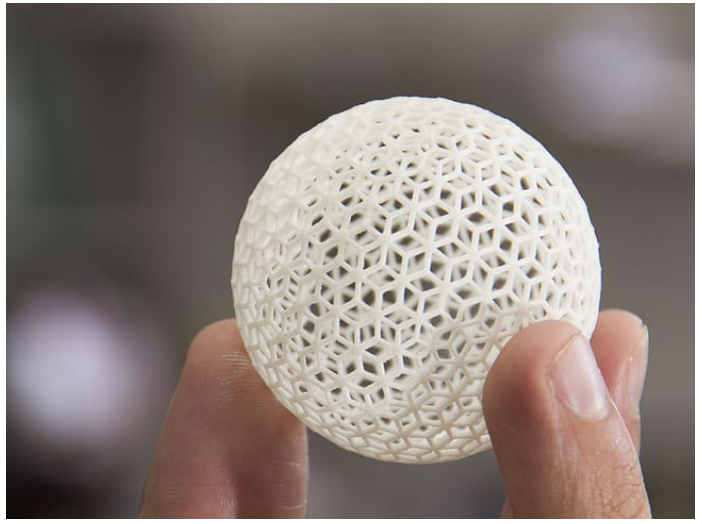
Fewer experiments to a good answer

Materials design is filled with difficult property tradeoff decisions

Strength vs ductility



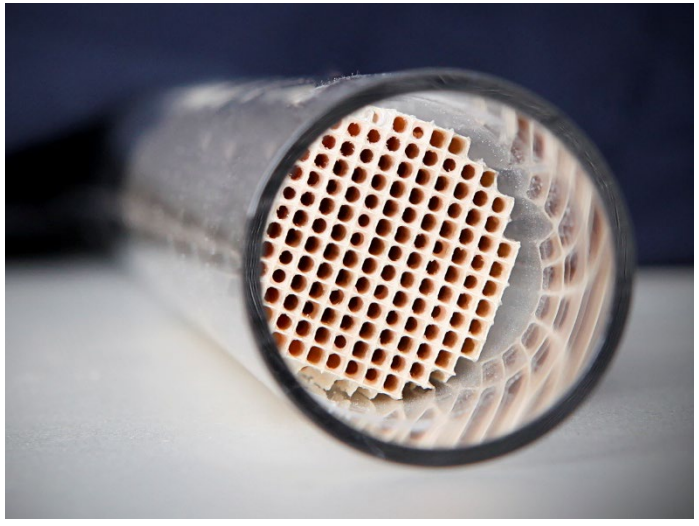
3d print parameters



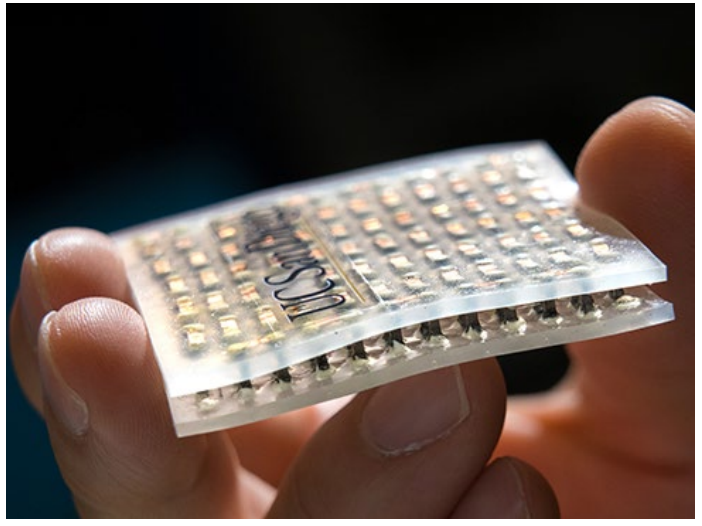
Performance vs biodegradability



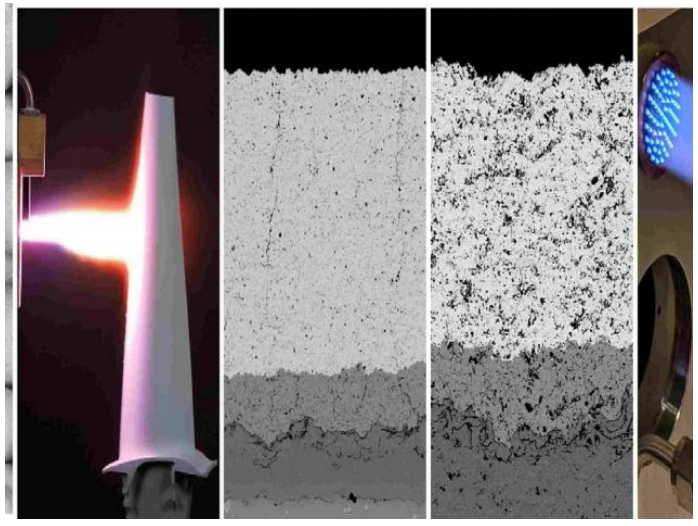
Catalytic activity vs selectivity



Thermal vs electrical



Thermal vs mechanical



A motivating example… let's get that 3d printer up and running!

Your company purchased a brand new 3D printer and wants to use it to fulfill a client request for a custom part. Your manager gives you a week to get the machine dialed in before they print the client's part.

What is the most efficient way to dial in these params?

- X-offset
- Y-offset
- Prime Delay
- Print Speed

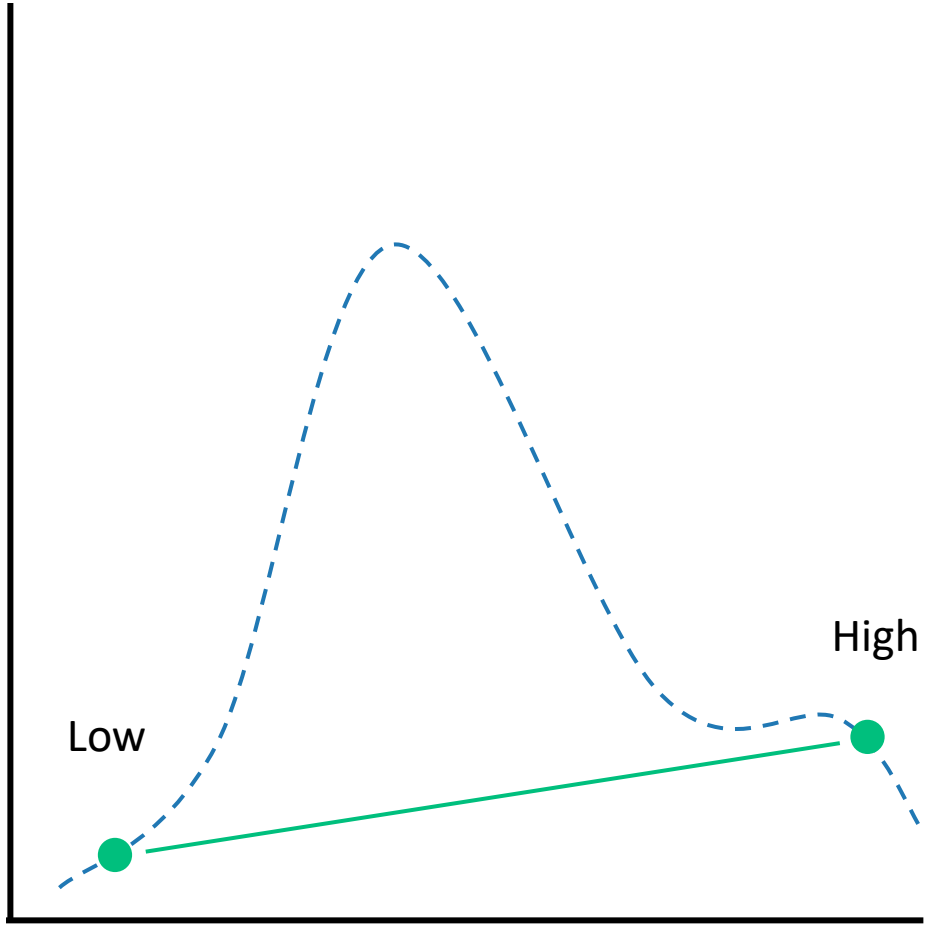
A traditional Design of Experiment (DOE) approach is too slow

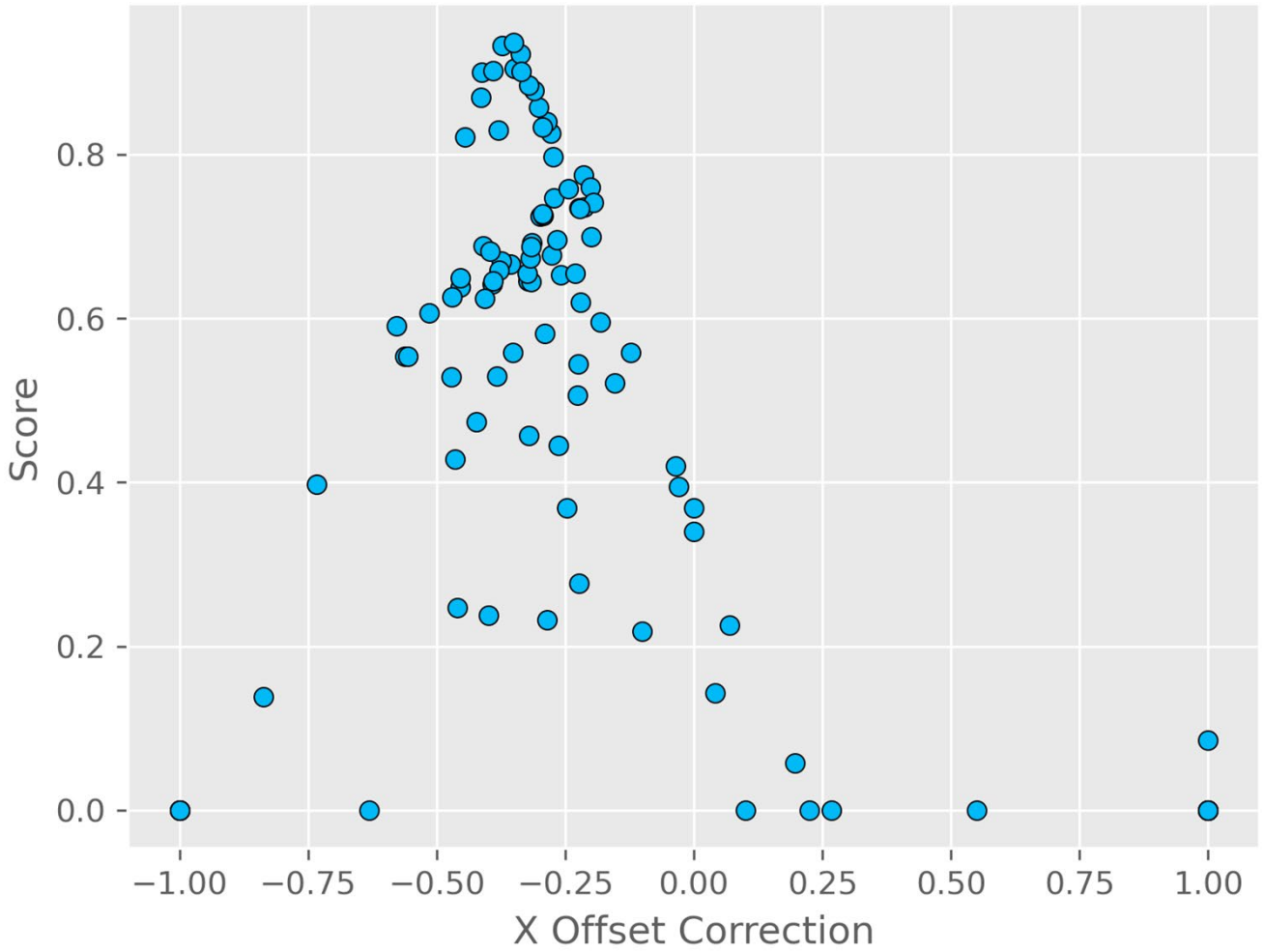
Create statistically orthogonal testing points to learn the impact of each parameter on the printed part properties.

For 2 level DOE approach with 3 replicates:

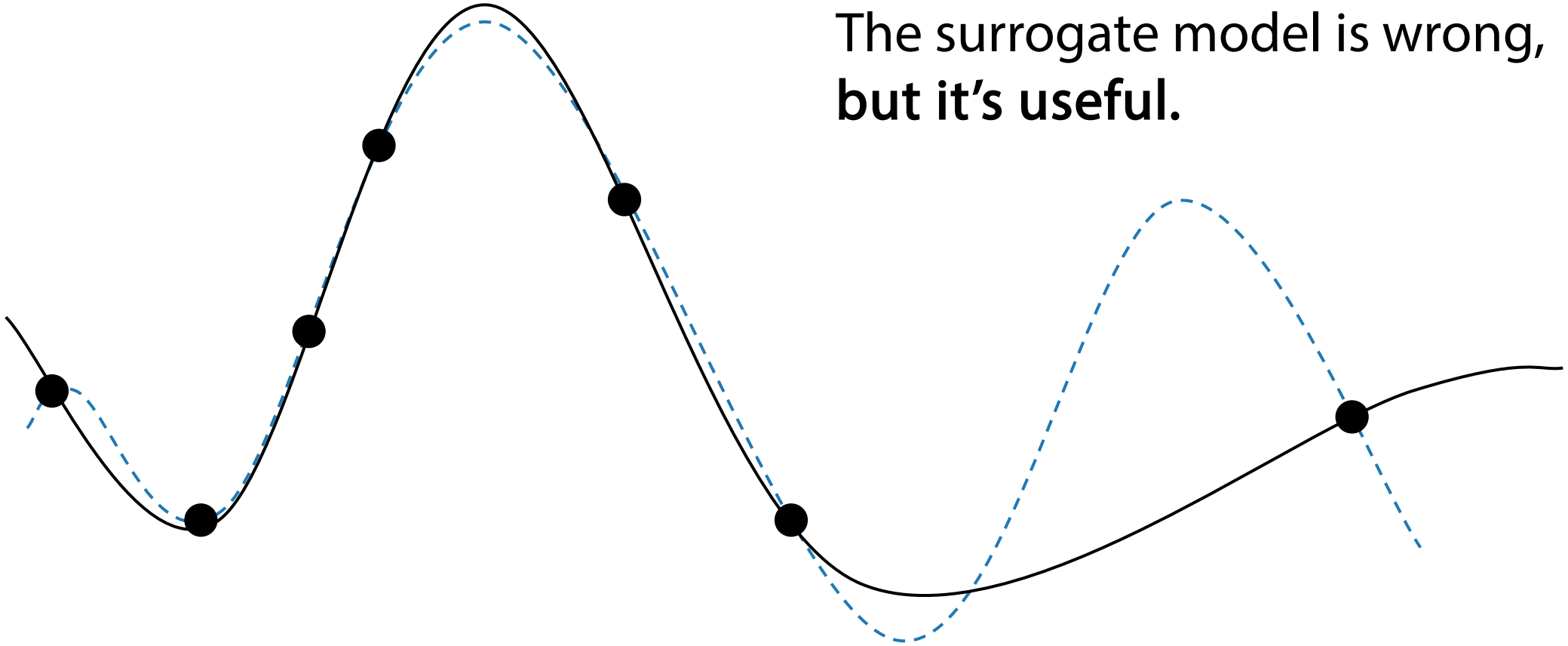
Full Factorial $3 * 2^4 = 48$ Experiments ← And that's just the beginning!

DOE isn't just slow though, it can miss optimal solutions!



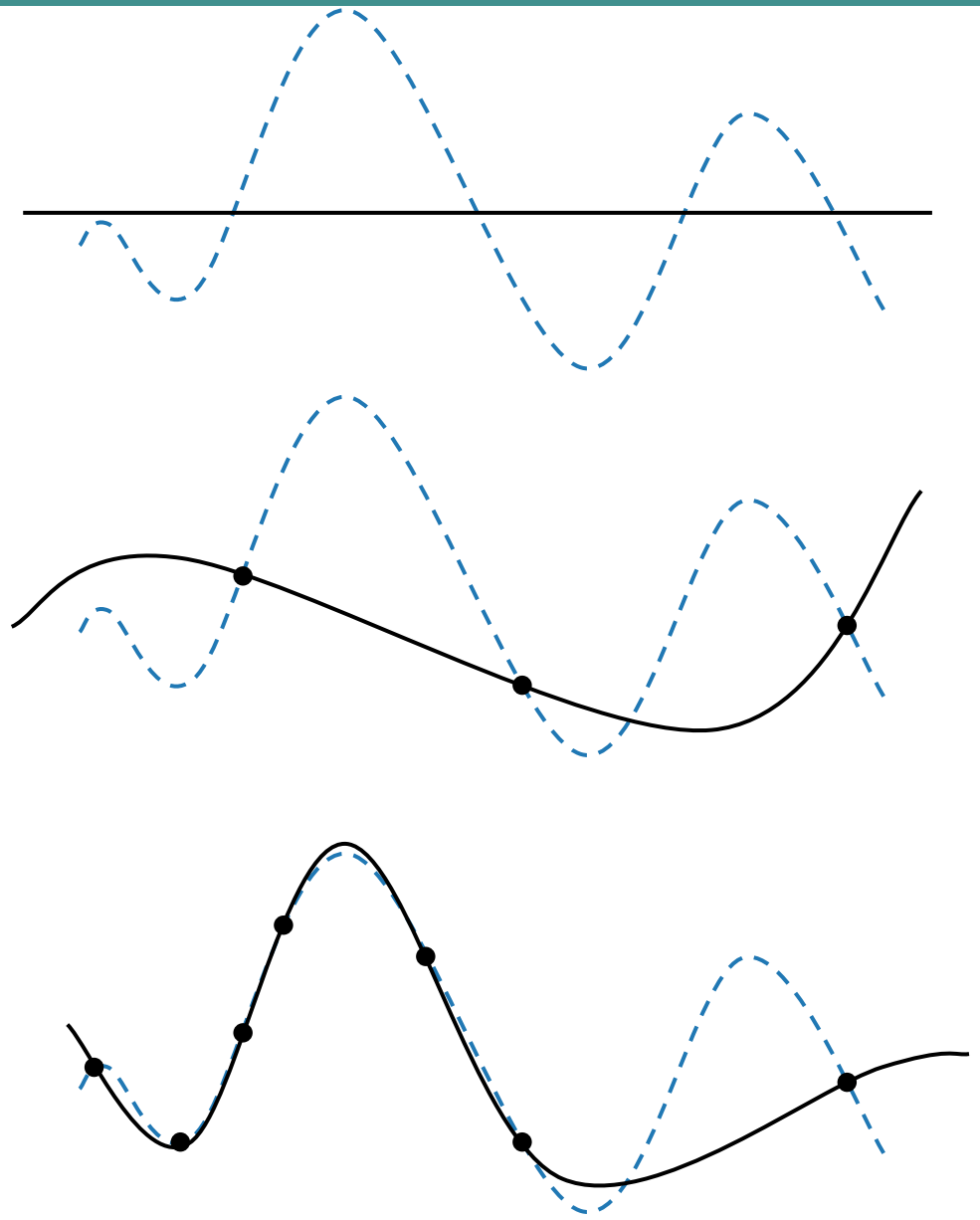


A surrogate model can help us understand the objective function



The surrogate model is wrong,
but it's useful.

We can improve our understanding with each new data point we collect



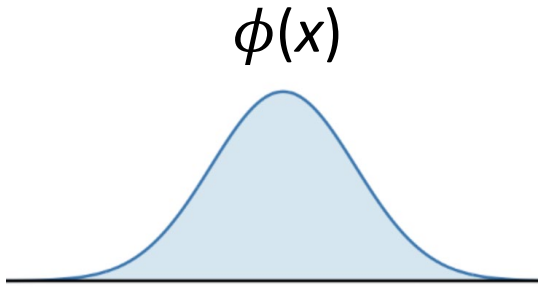
Our initial assumption about the objective function

Observing some points gives the function some form.

Our belief about that form changes with more observations.

Bayesian: start with a belief, then update it with every result

Consider a situation where we want to model a value y with a distribution ϕ with parameters x



Prior: What do we believe about ϕ before observing data.

Likelihood: Where is ϕ most compatible with the observation y

Posterior: What is the distribution of ϕ having observed y.

$$p(\phi \mid x, y) = \frac{p(\phi \mid x)\, p(y \mid x, \phi)}{p(y \mid x)}.$$

Prior × likelihood = posterior: data narrows what we believe

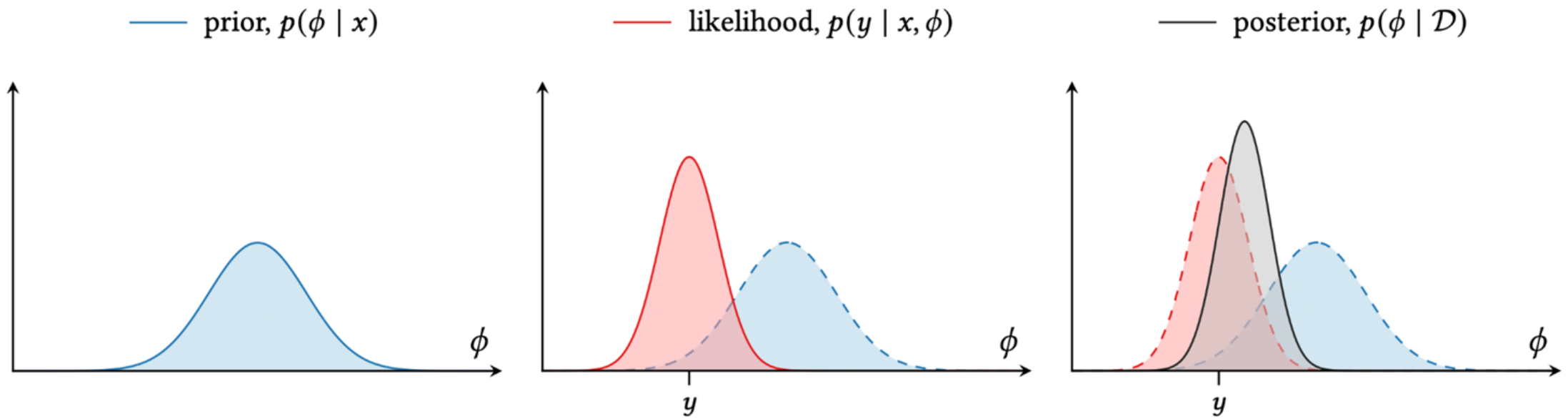
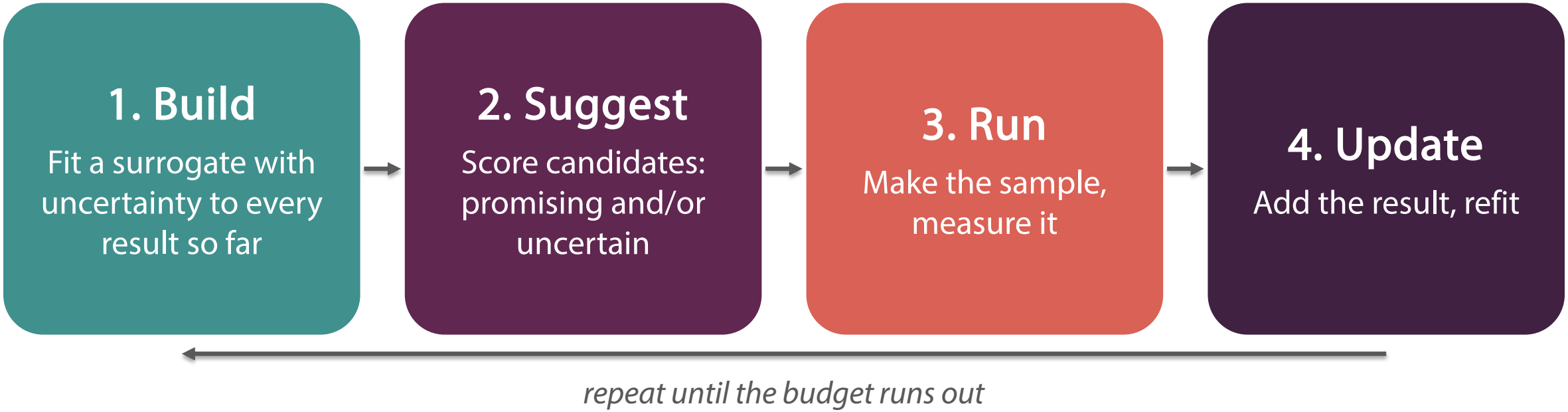


Figure Referenced From

Garnett, Roman. Bayesian Optimization. 2023. Cambridge University Press

The loop: build a model, suggest an experiment, update

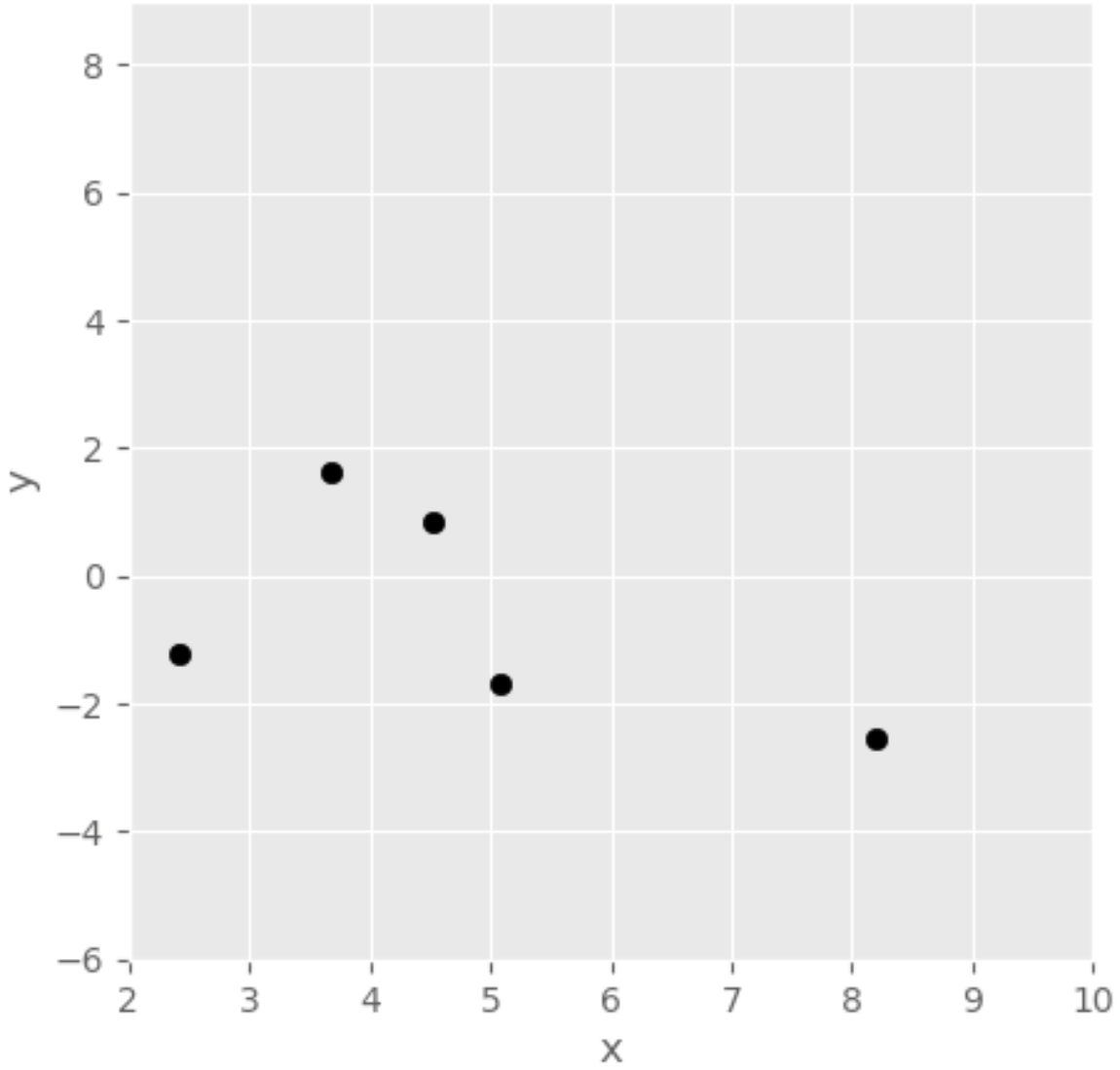


The model doesn't need to be right; it needs to rank where to look next

Gaussian processes

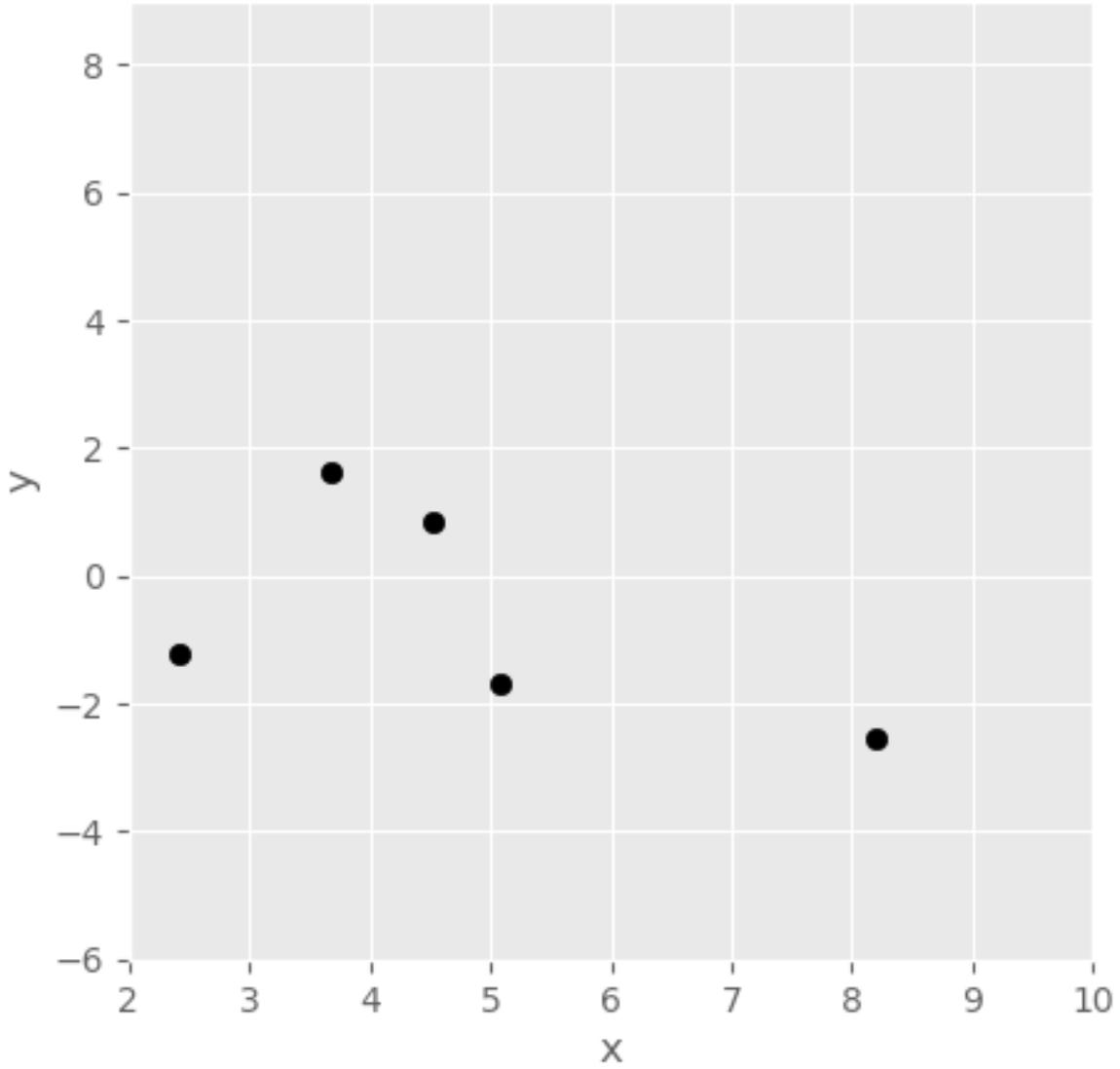
A surrogate that knows what it doesn't know

Why should we care about this?

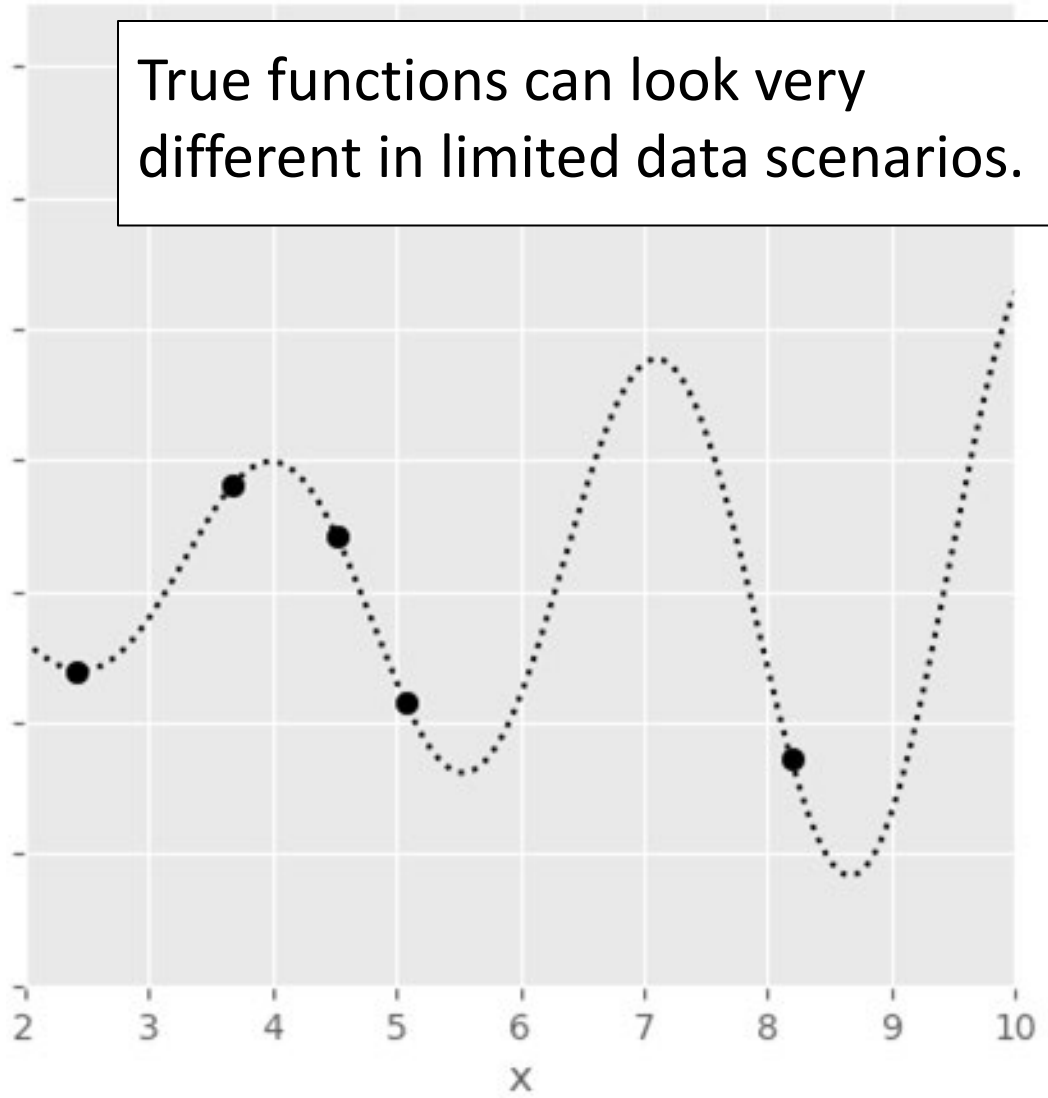


How would you fit this function?

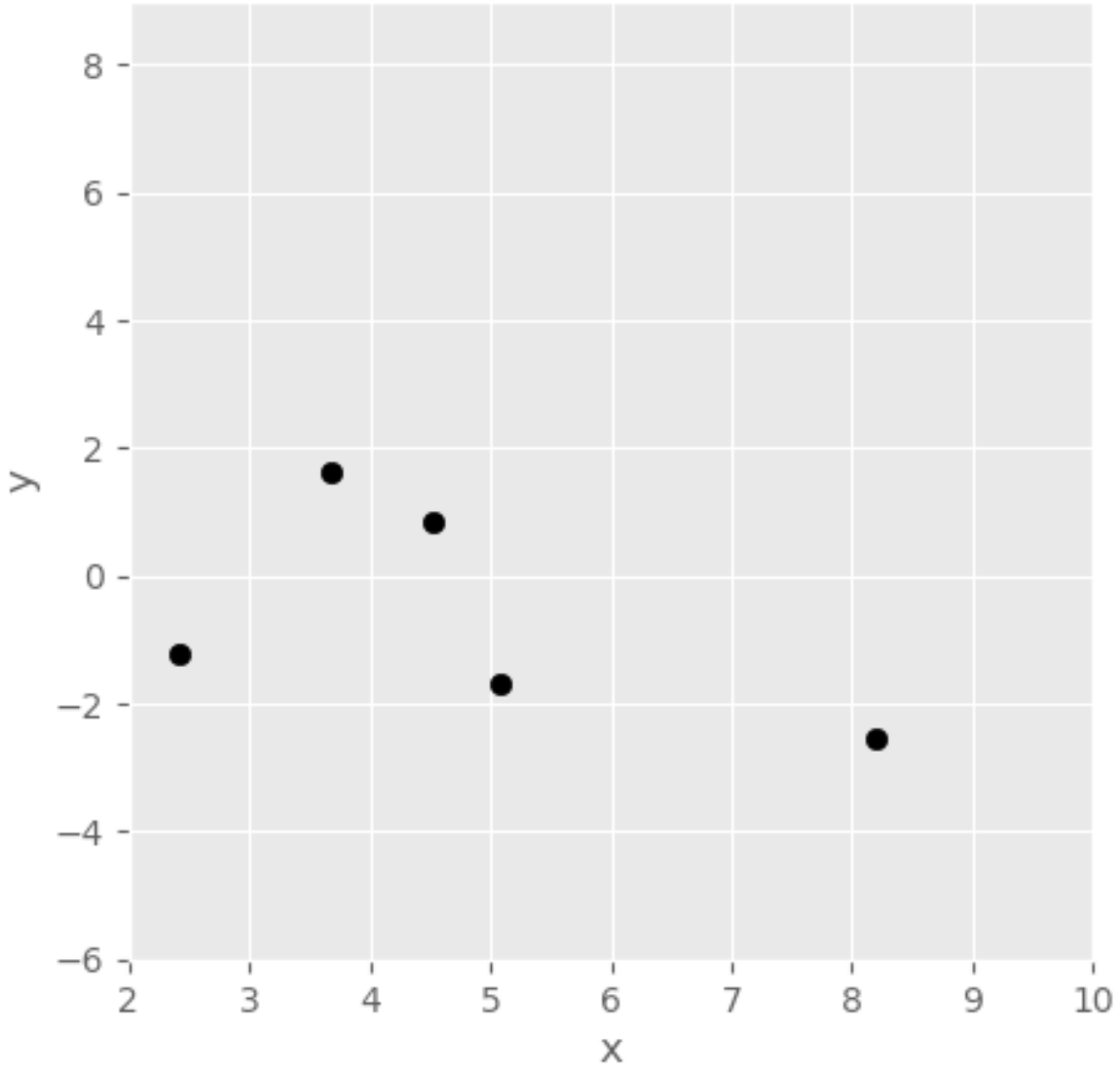
With little data, very different functions fit equally well



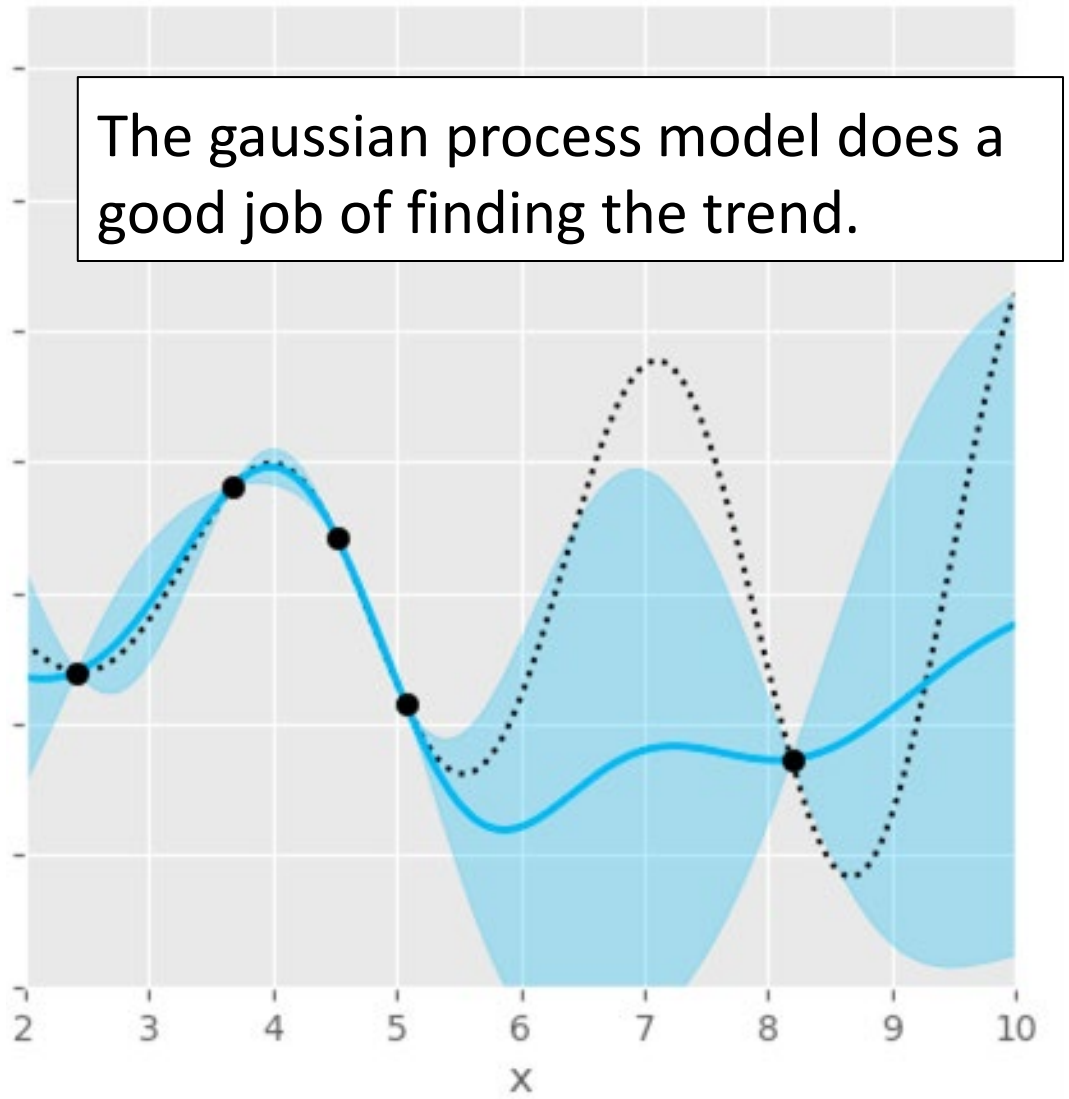
True functions can look very different in limited data scenarios.



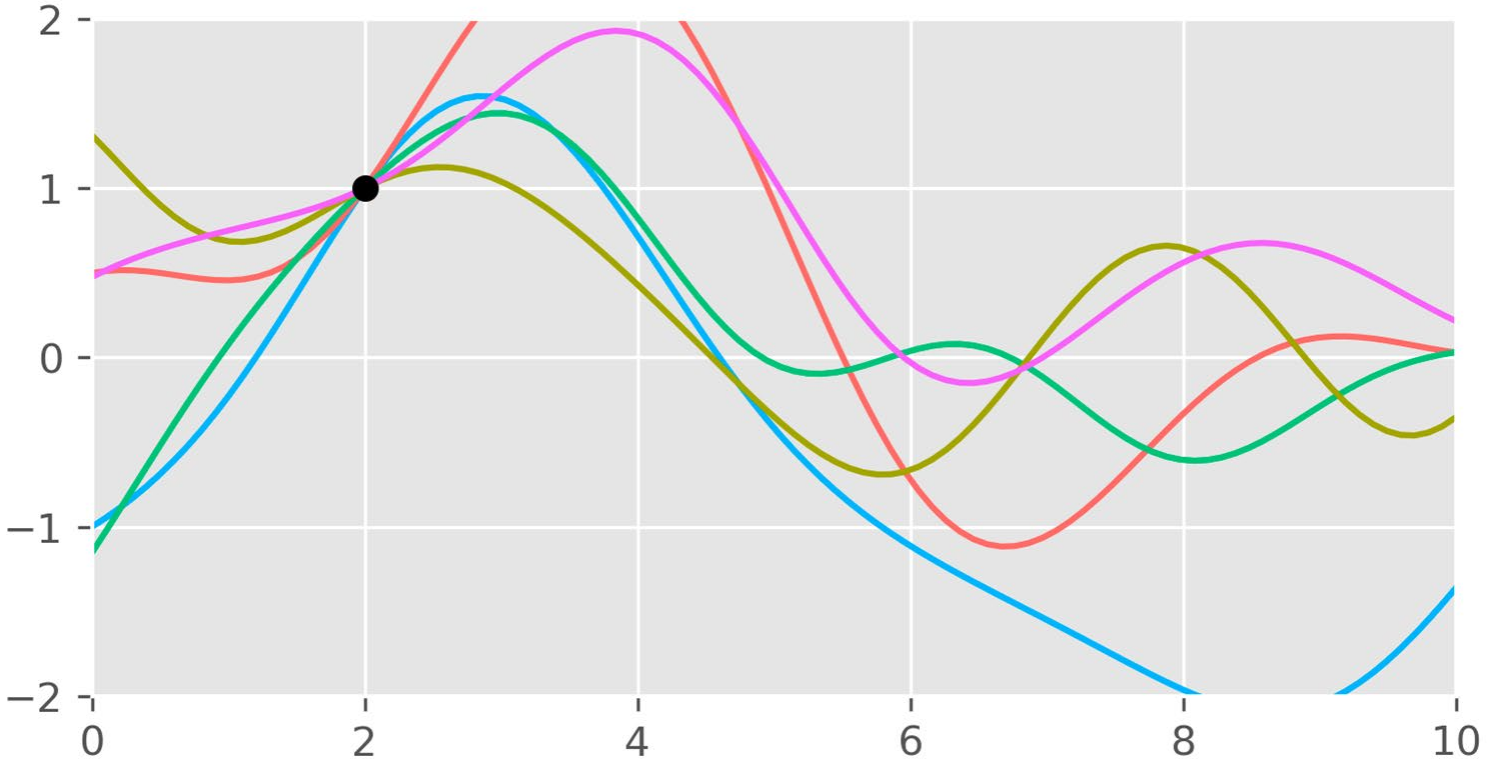
A GP gives a best guess and an honest band of uncertainty



The gaussian process model does a good job of finding the trend.

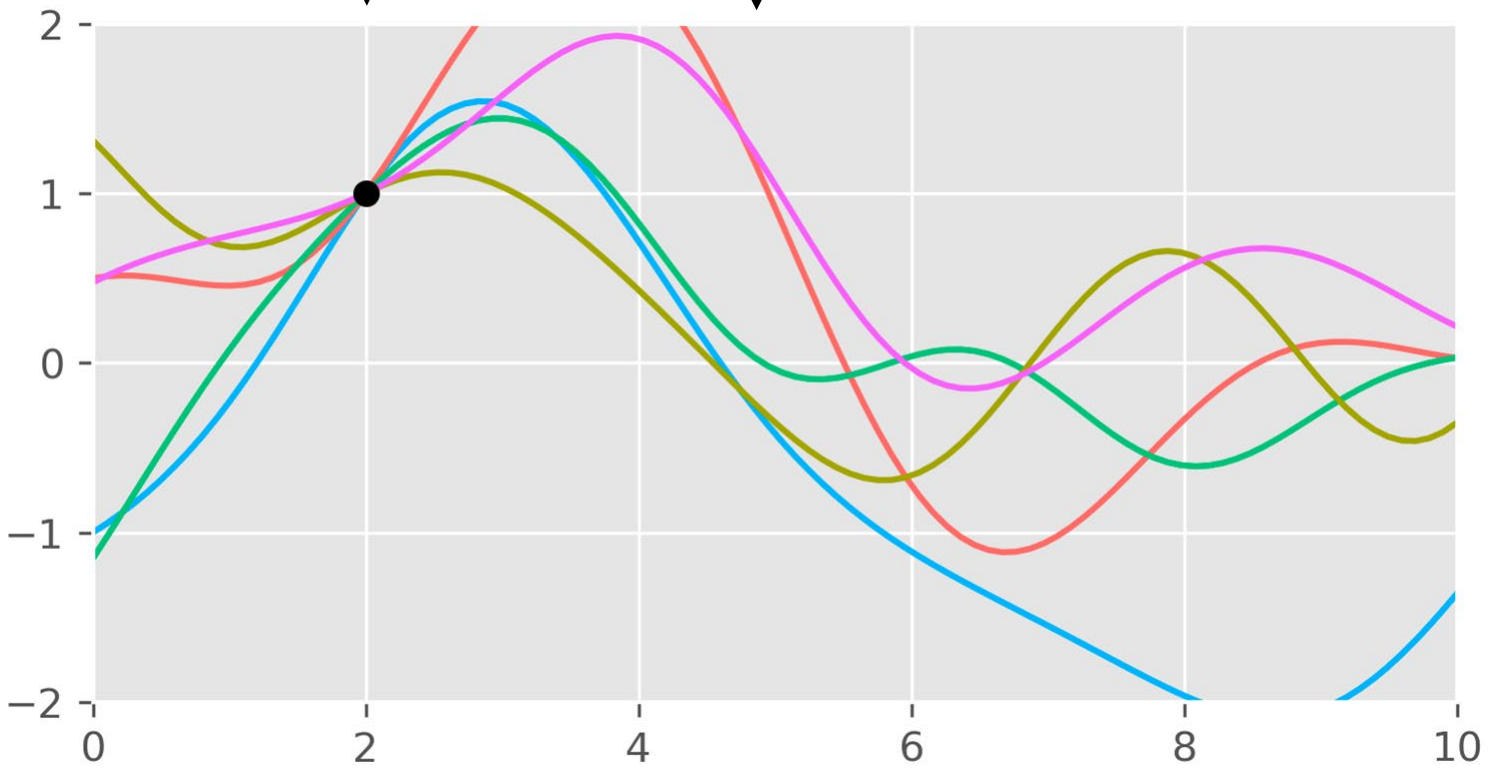


Function draws can then be conditioned on observed data!



Conditioning on data keeps only the functions that fit

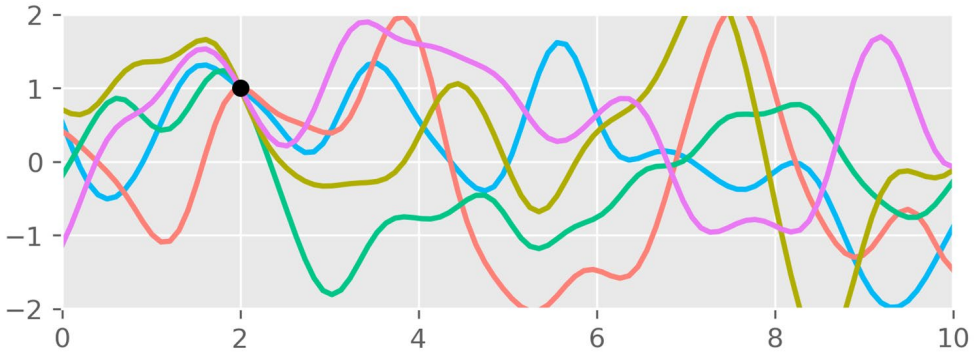
How much should this observation influence this point?



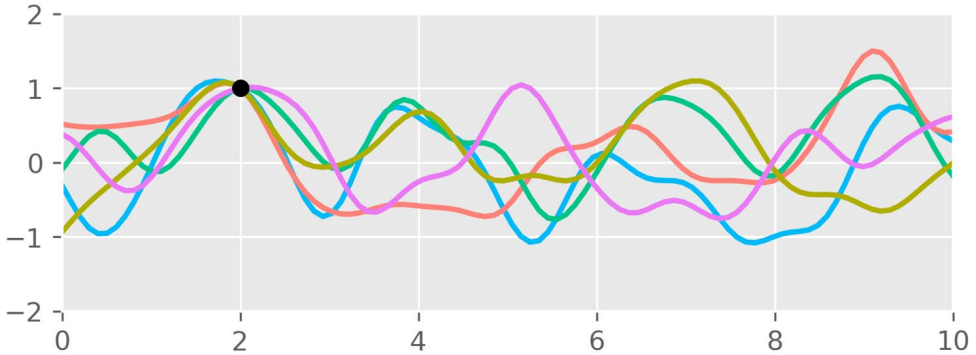


Correlation (lengthscale)

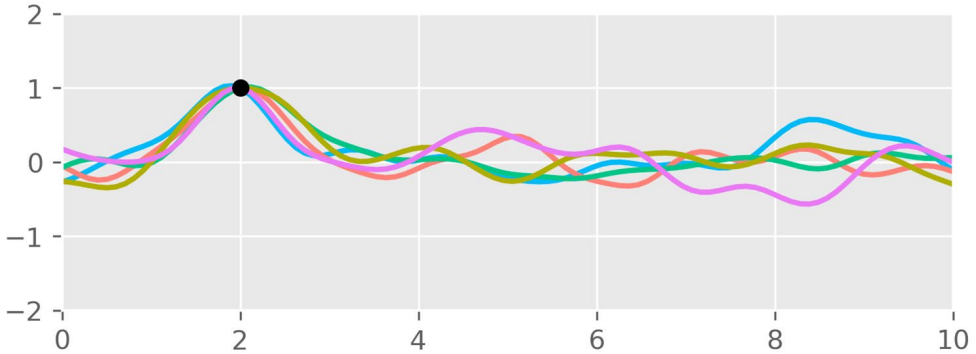
The variance parameter controls the spread of possible functions



$\sigma = 1.0$



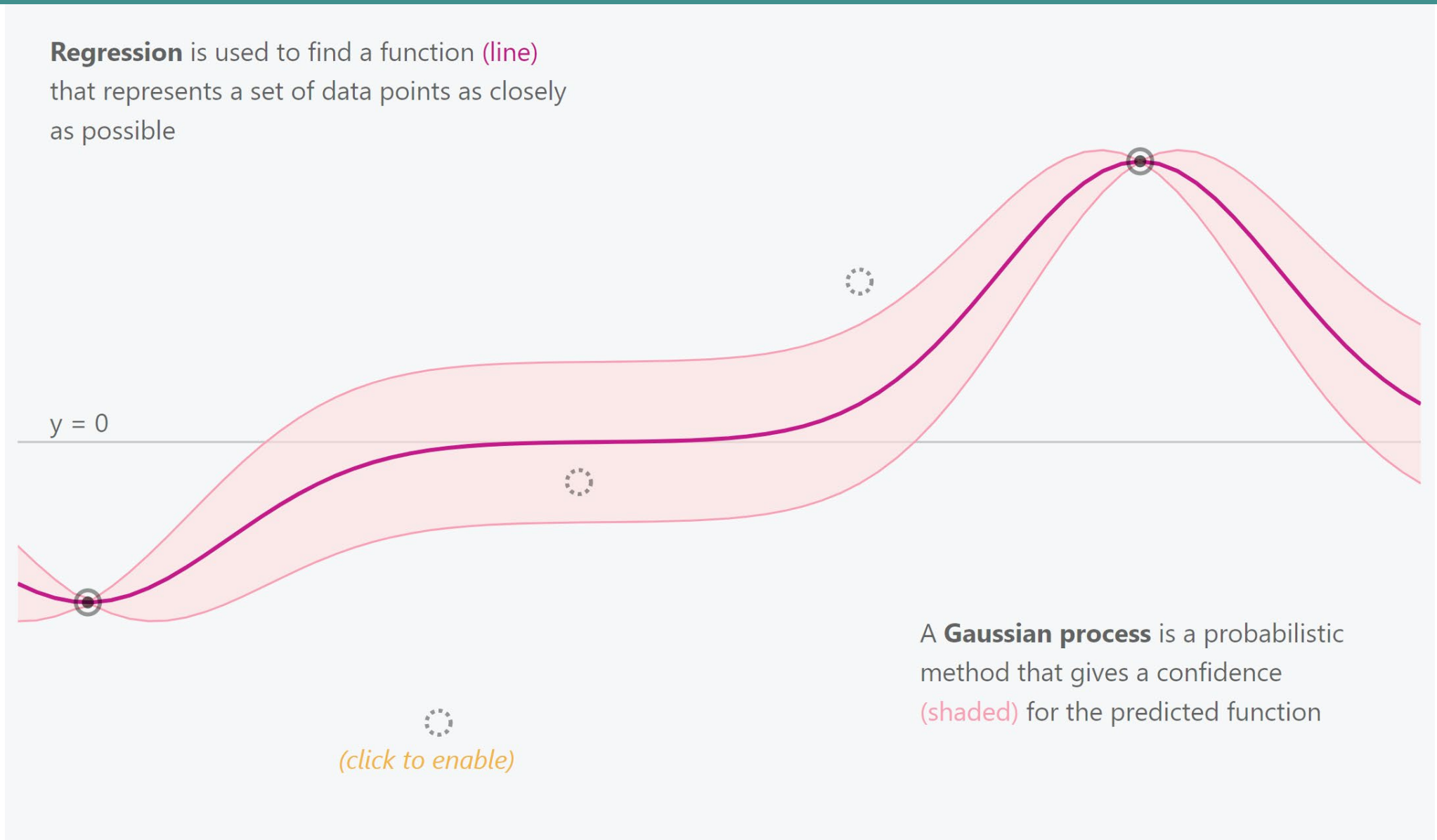
$\sigma = 0.25$



$\sigma = 0.05$

http://www.infinitecuriosity.org/vizgp/

Far from data, a GP falls back to the mean and widens its band



https://distill.pub/2019/visual-exploration-gaussian-processes/

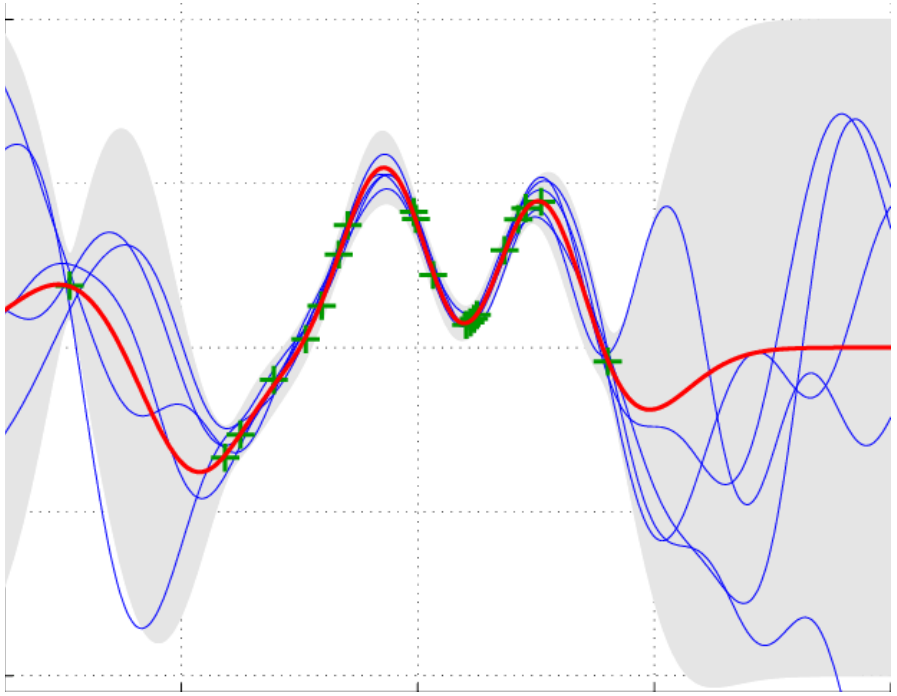
GPs struggle with many dimensions and extrapolation

Extrapolation does really poorly with Gaussian processes (depends on length scale of kernel)

They are not sparse (they use the whole samples/features information to perform the prediction

They lose efficiency in high-dimensional spaces (features > a few dozen)

Picking and tuning kernel is tricky

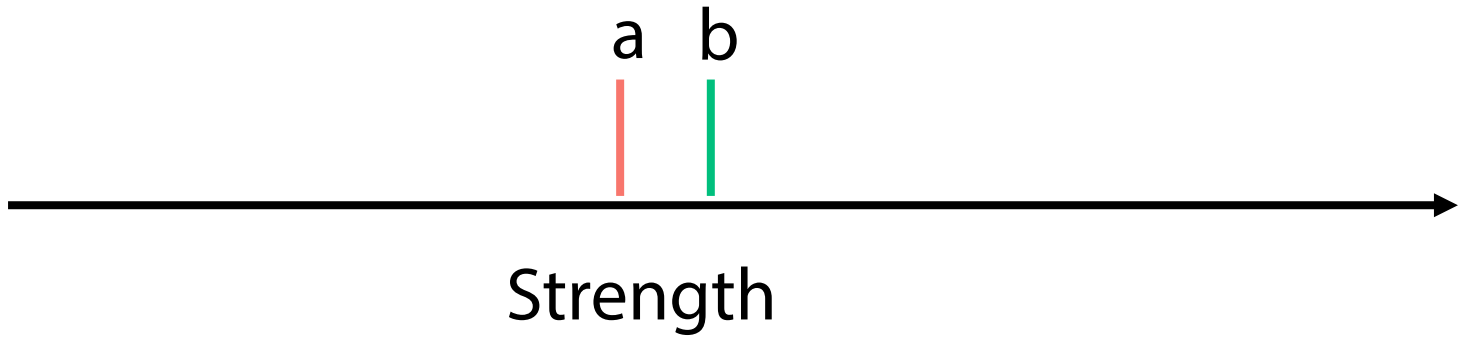


Perez-Cruz et al Gaussian Processes for Nonlinear Signal Processing: An Overview of Recent Advances IEEE

Choosing the next experiment

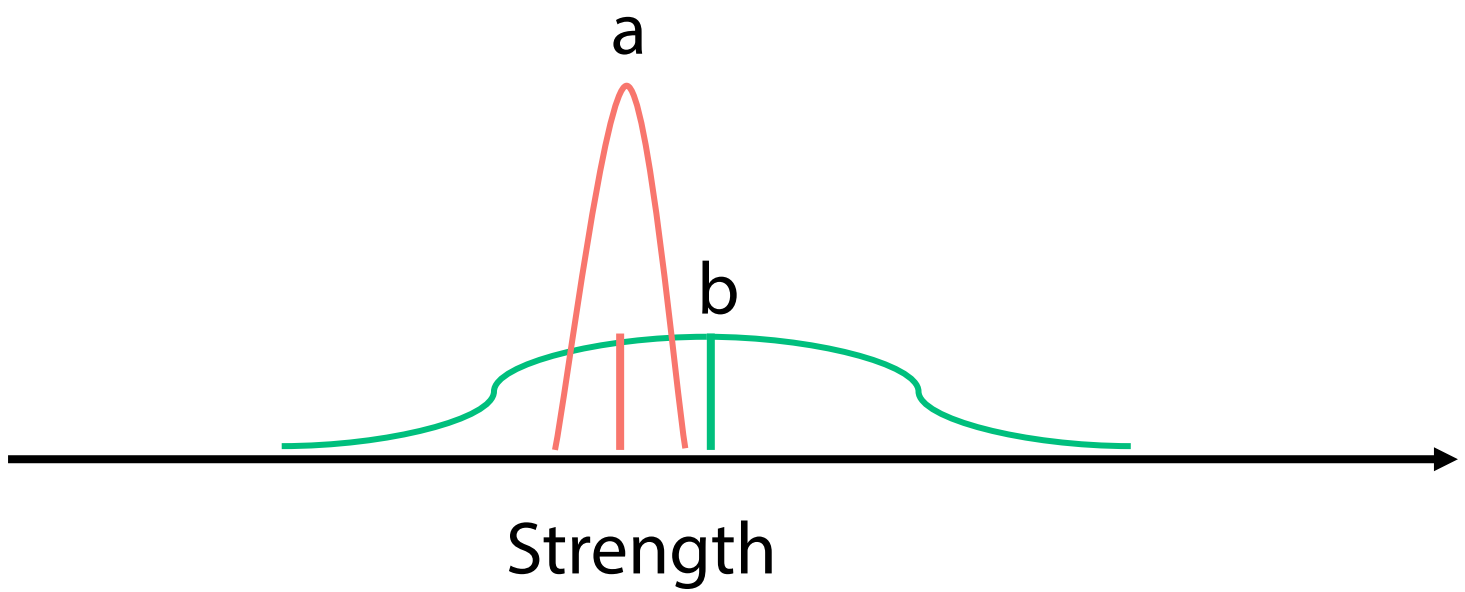
Exploit what you know, explore what you don't

Uncertainty changes the decision-making process



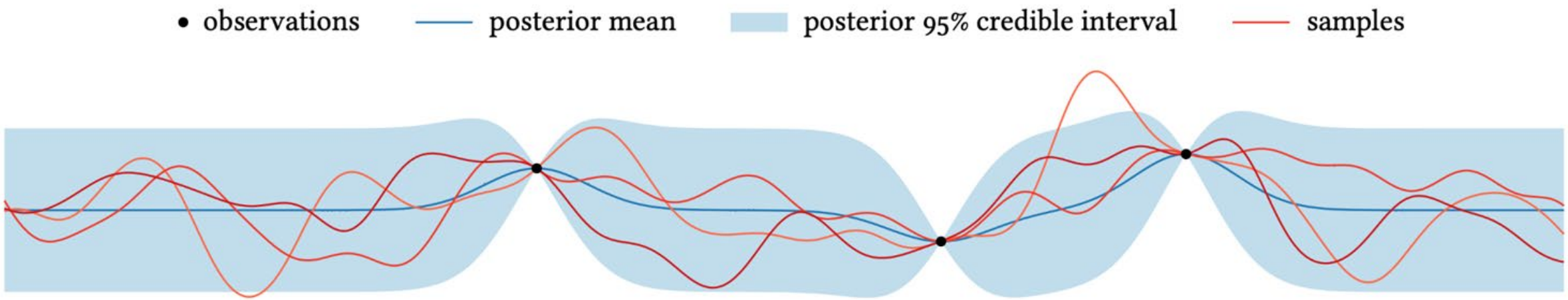
Which material would you choose for a bridge support?

Same mean, different spread: the right pick depends on the goal



Which material would you choose for a bridge support?

Where would you run the next experiment, and why?



Where would you look next? why?

Figure Referenced From

Garnett, Roman. Bayesian Optimization. 2023. Cambridge University Press

Every pick trades exploring the unknown against exploiting the best guess

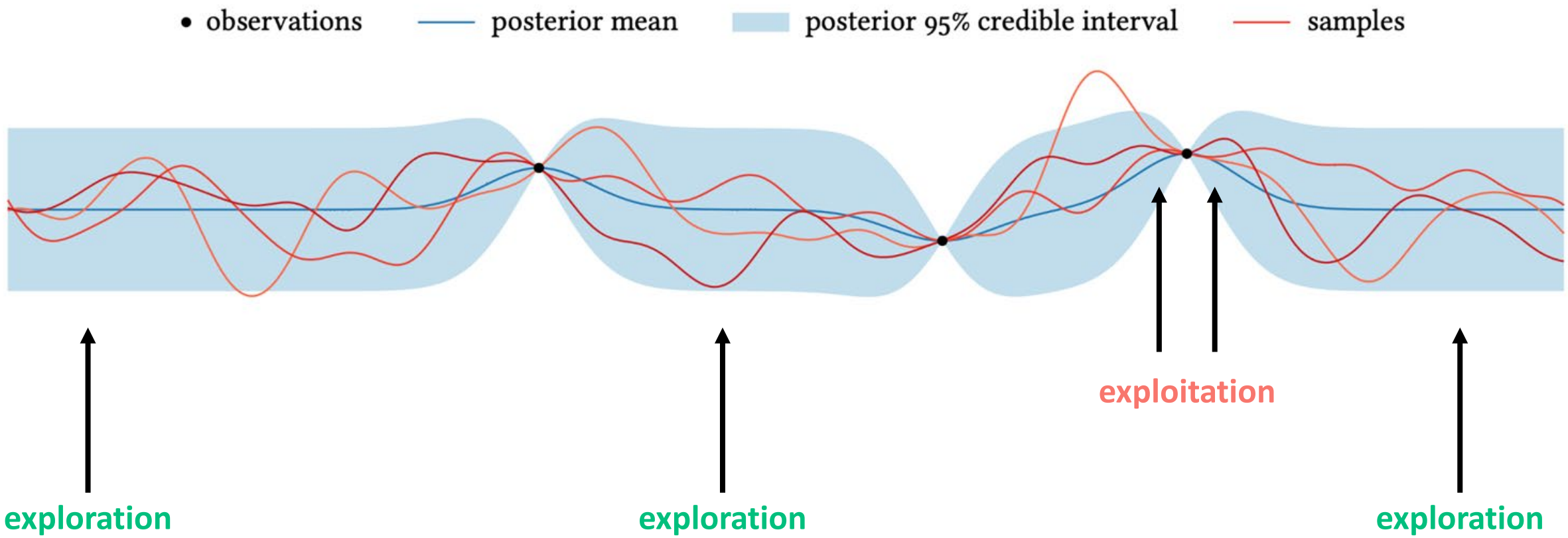
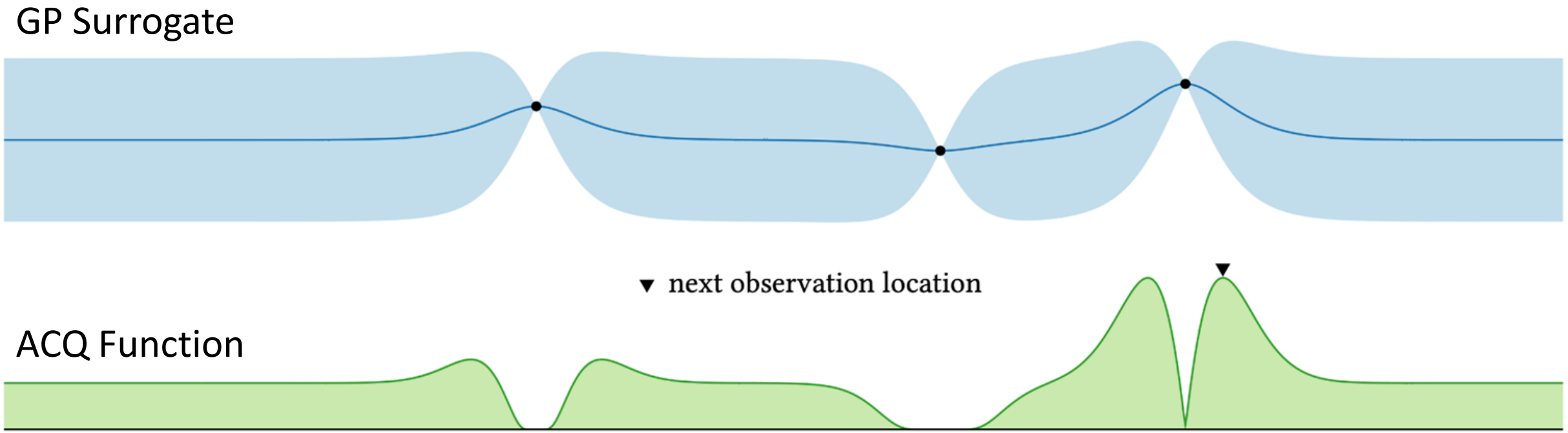


Figure Referenced From

Garnett, Roman. Bayesian Optimization. 2023. Cambridge University Press

An acquisition function scores every candidate; run the top score next



The acquisition function applies preference to the value of our surrogate model at any given point.

Figure Referenced From
Garnett, Roman. Bayesian Optimization. 2023. Cambridge University Press

UCB = mean + λ × uncertainty: a larger λ explores more

$$a(x;\lambda) = \mu(x) + \lambda\sigma(x)$$

Mean + Uncertainty

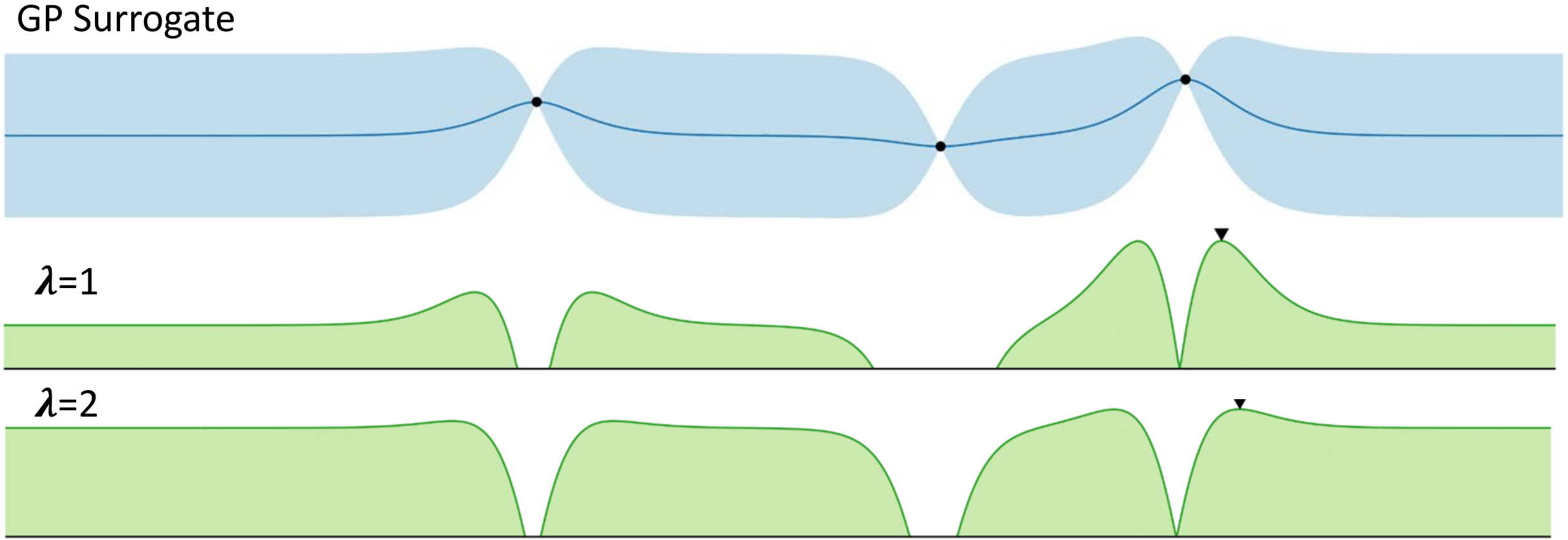


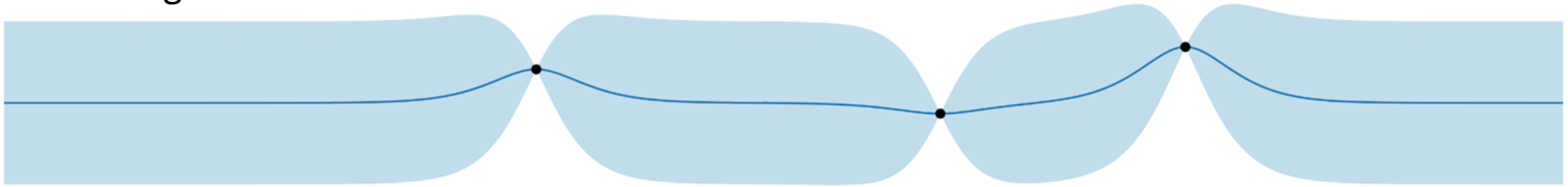
Figure Referenced From: *Garnett, Roman. Bayesian Optimization. 2023. Cambridge University Press*

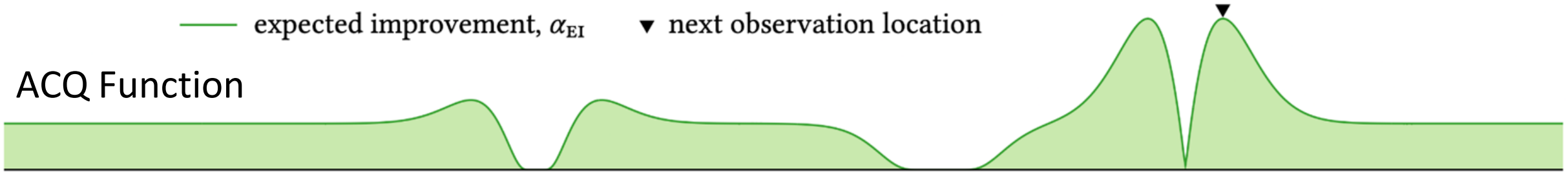
Expected improvement: how much might we beat our best result?

$$\mathrm{EI}(x) = \left(\mu - f(x^\star)\right)\Phi\left(\frac{\mu - f(x^\star)}{\sigma}\right) + \sigma\varphi\left(\frac{\mu - f(x^\star)}{\sigma}\right)$$

Mean + Uncertainty

GP Surrogate





ACQ Function

Figure Referenced From: *Garnett, Roman. Bayesian Optimization. 2023. Cambridge University Press*

The acquisition function is just an opinion, and there are many

Probability of Improvement
Knowledge Gradient
Mutual Information
Thompson Sampling
Entropy Search
Max-Value Entropy Search
…

Acquisition functions turn μ and σ into a decision

Upper confidence bound

$UCB = \mu + \lambda\sigma$

$\lambda = 0$: pure exploitation. Large λ: exploration

Expected improvement

How much do I expect to beat my best?

ξ adds an exploration margin

Uncertainty sampling

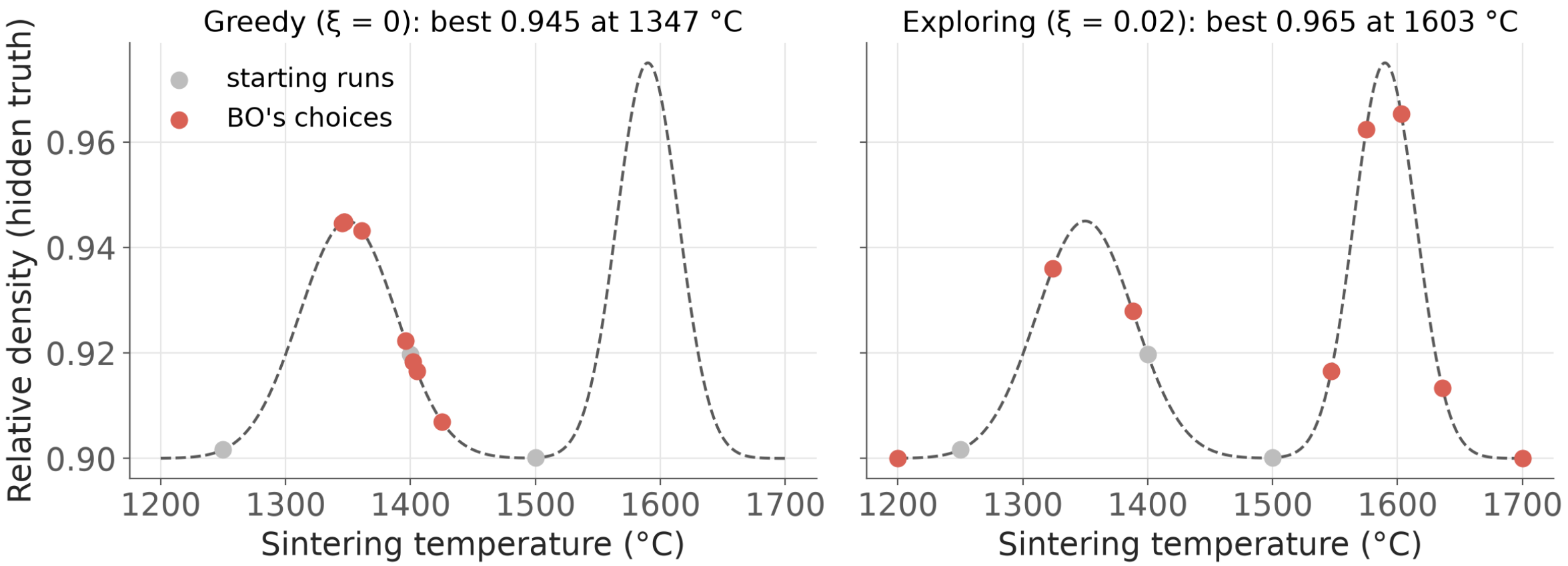
Pick the largest σ

Active learning: build the best model, not the best sample

Every experiment either exploits or explores. BO makes the trade-off explicit

Greedy search got stuck; a nudge to explore found the better peak

Simulated sintering: 3 starting runs + 8 suggestions; the optimizer never sees the curve



Simulated response made for this course (notebook Part A).

Your starting data matters: for the same budget, random beats a grid

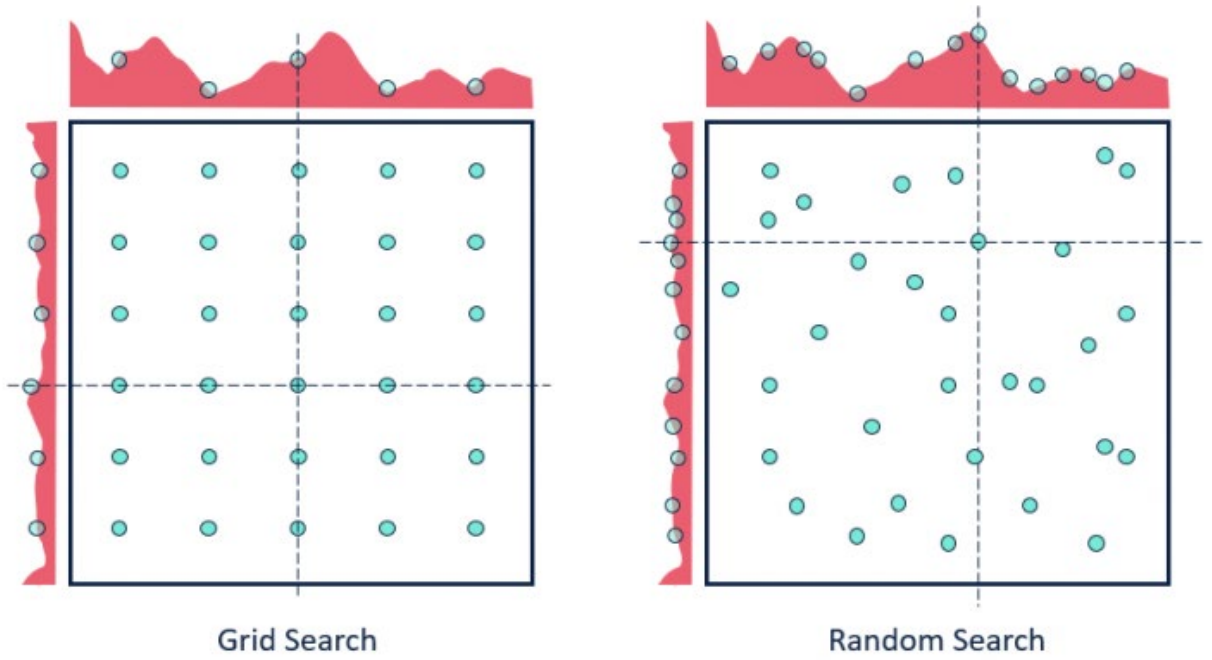
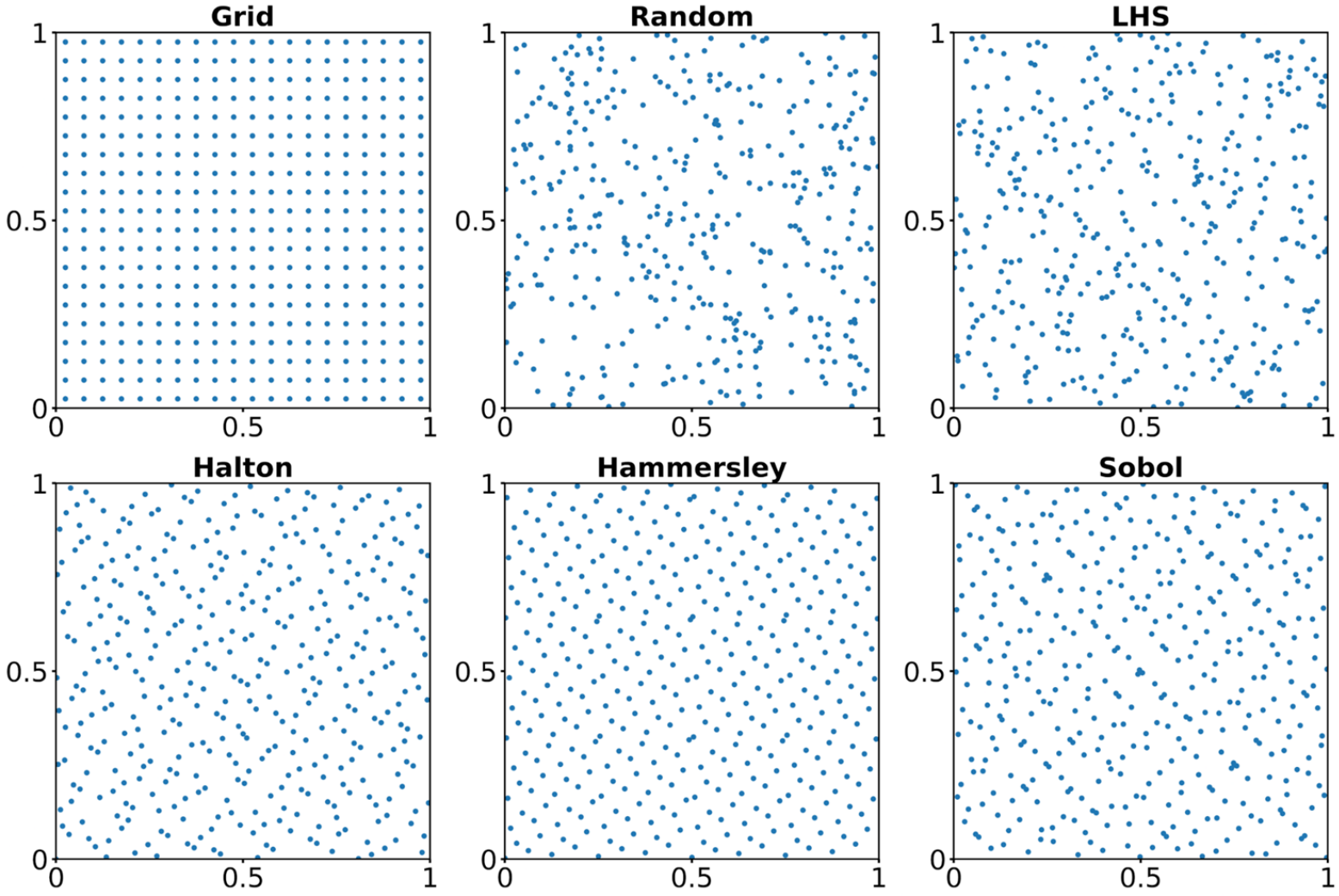


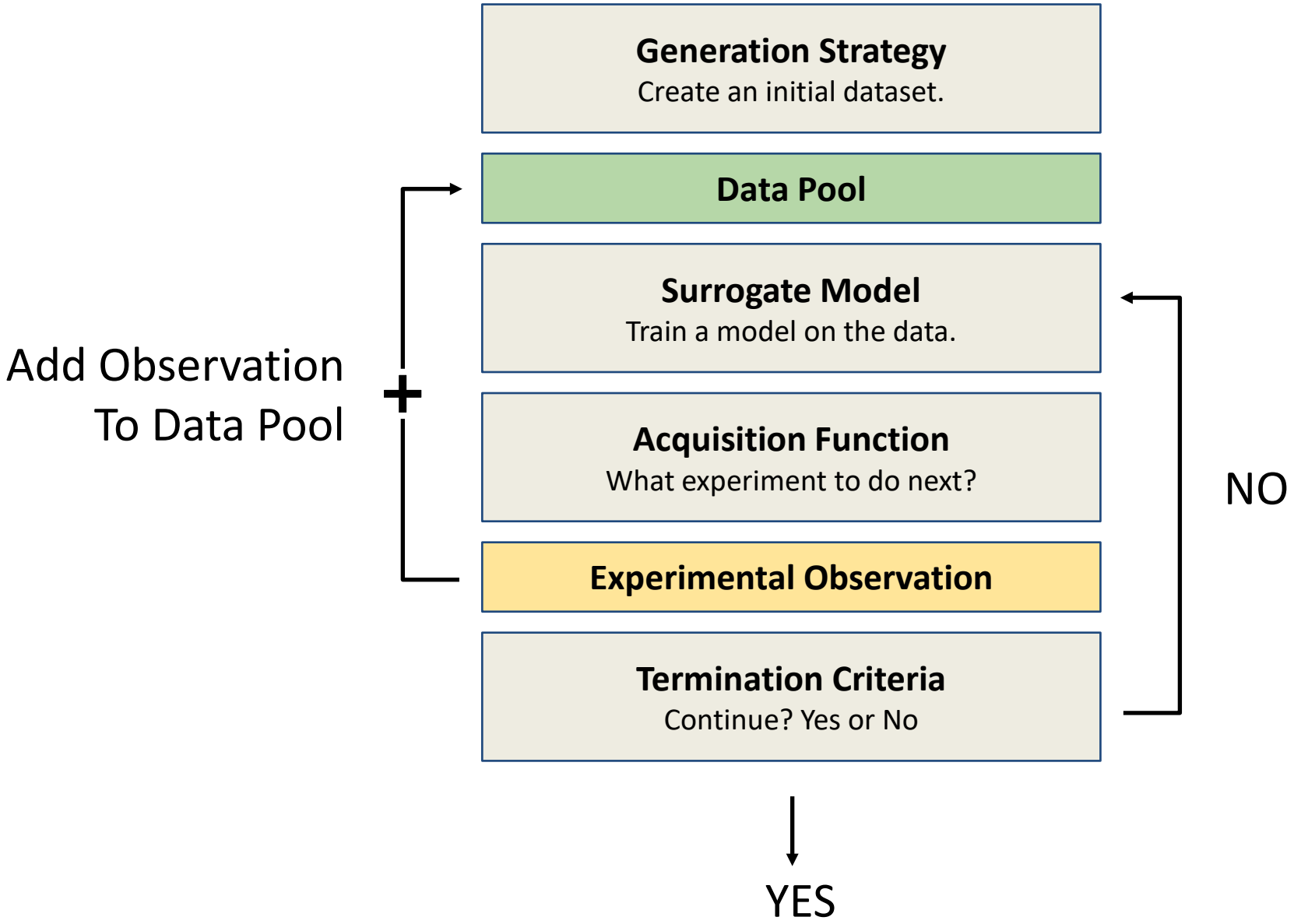
Figure Referenced From:
https://larevueia.fr/3-methodes-pour-optimiser-les-hyperparametres-de-vos-modeles-de-machine-learning/

Quasi-random sequences fill the space evenly without a grid



Wu C, et al. A comprehensive study of non-adaptive and residual-based adaptive sampling for physics-informed neural networks. Computer Methods in Applied Mechanics and Engineering. 2023

Four blocks: starting data, surrogate, acquisition function, stopping rule

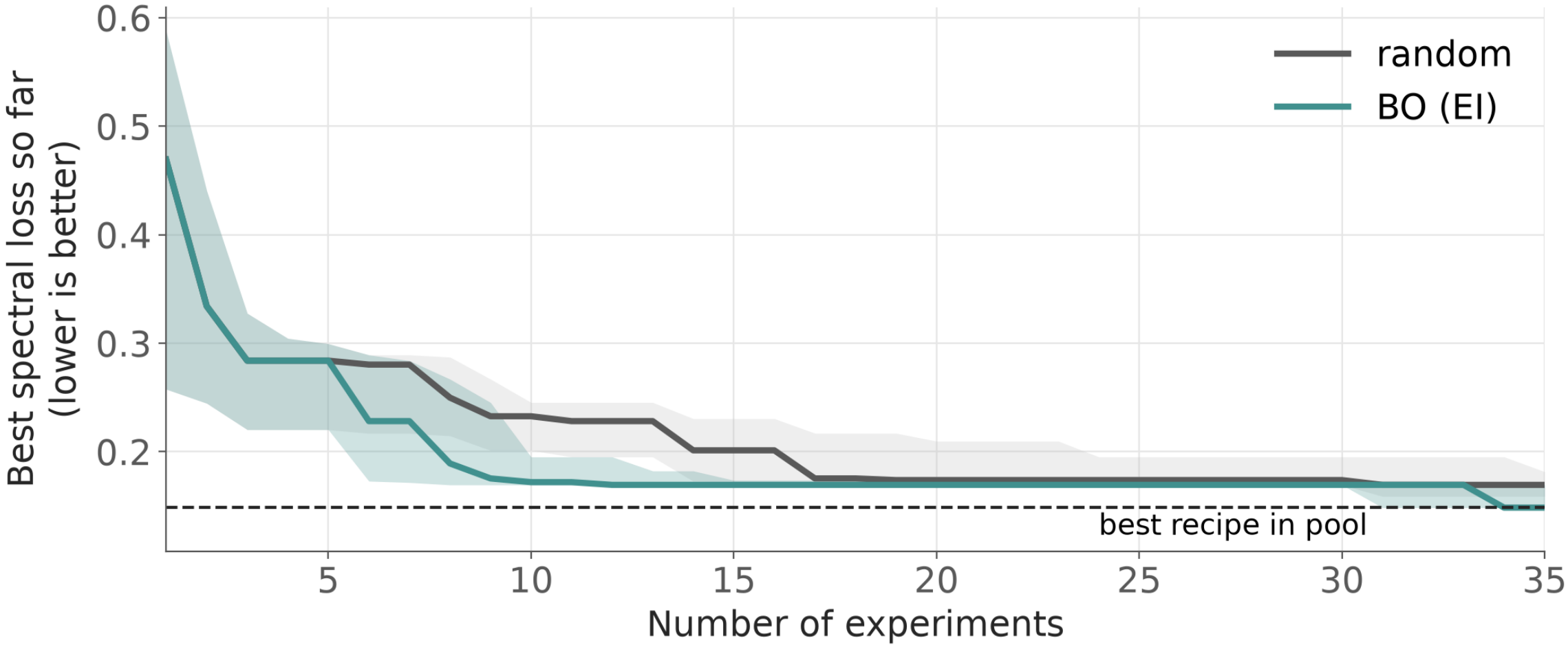


Does it work on real experiments?

Silver nanoparticles, thermoelectrics and polymer films

On real flow-synthesis data, BO reached a top-5% recipe in 6 runs vs 9

164 measured Ag nanoparticle recipes, 5 inputs; 15 campaigns per strategy (median, interquartile band)



Data: Mekki-Berrada et al., npj Comput. Mater. 2021, via Liang et al. 2021. Analysis made for this course.

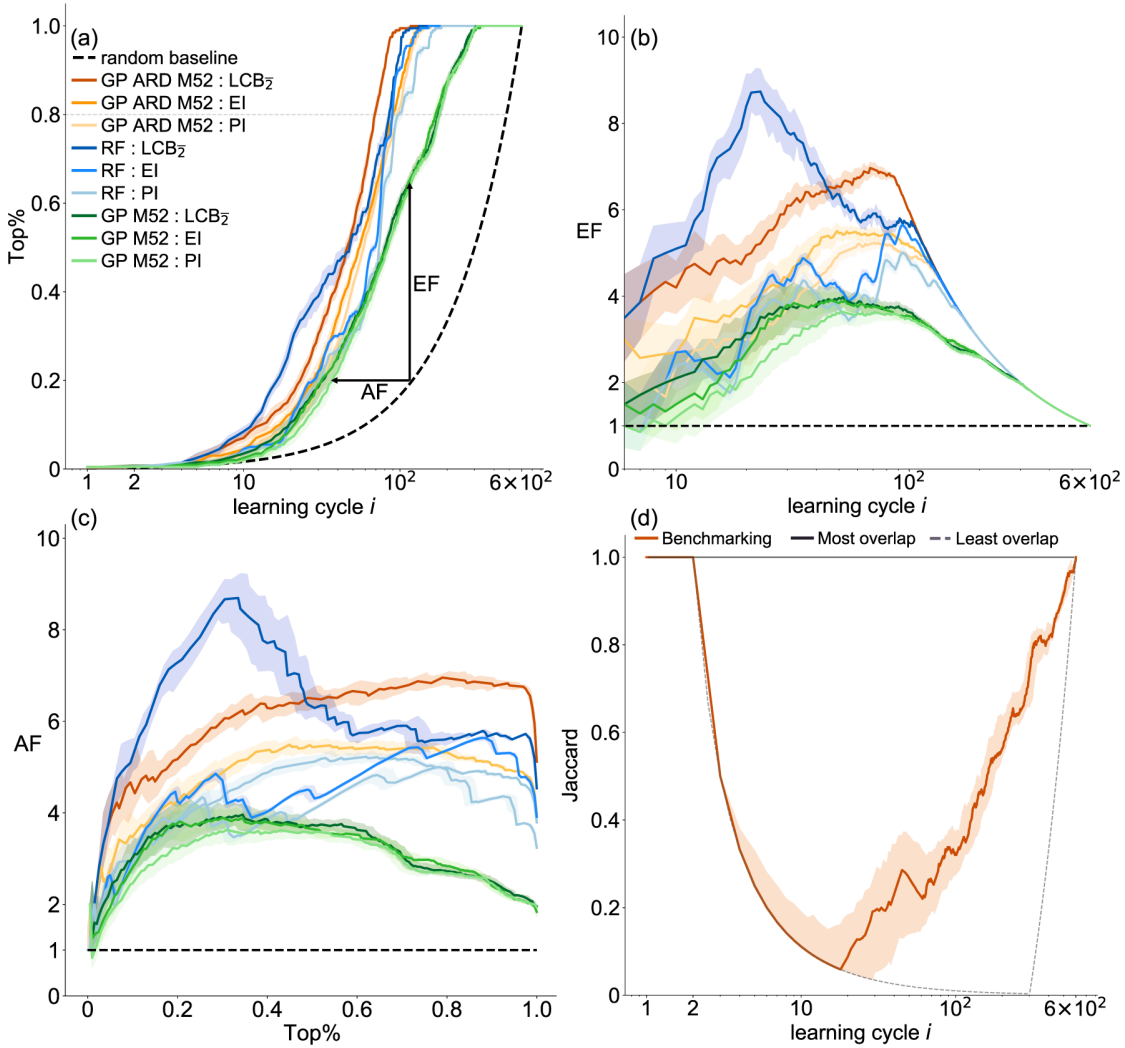
BO finds the best structures far faster than random sampling

GP-ARD and RF found 80% of the top-5% structures in about 75-85 of 600 tests; isotropic GP needed 170

Plain GP (isotropic)

One length scale for every input, so a knob that barely matters counts as much as a key one.

In the figure: GP M52



GP-ARD

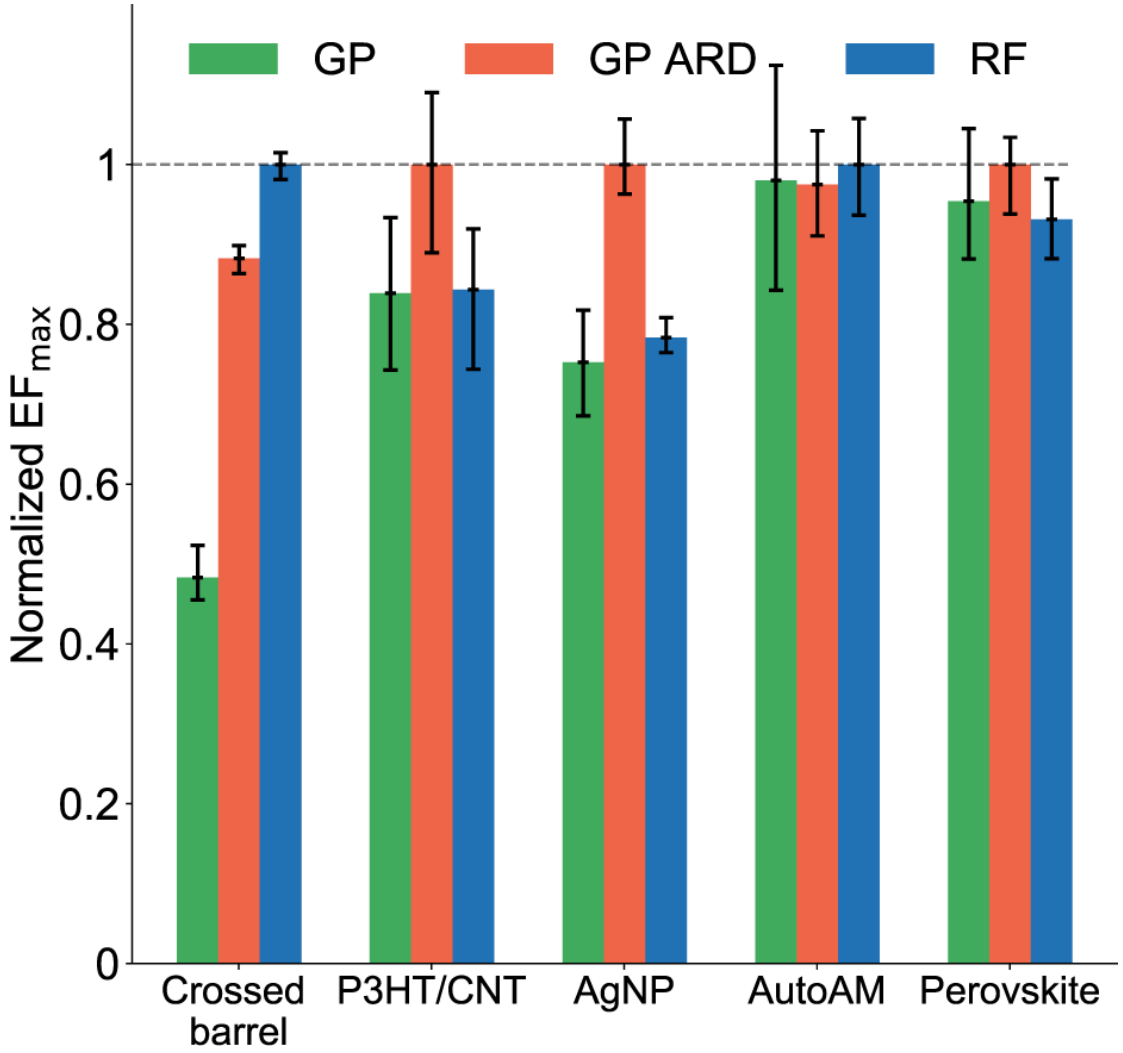
Its own length scale for each input, fitted from the data. A long one means that knob barely matters.

In the figure: GP ARD M52

Liang et al., npj Comput. Mater. 7, 188 (2021) https://doi.org/10.1038/s41524-021-00656-9

Anisotropic GP and random forest are the safest surrogates

Absolute EFmax was 2x-4x on four datasets and up to 8x on the grid-sampled Crossed barrel set



Liang et al., npj Comput. Mater. 7, 188 (2021) https://doi.org/10.1038/s41524-021-00656-9

A thermoelectric campaign: 612 p-type compositions to choose from

Goal

High power factor $S^2\sigma$ near room temperature, p-type

Features

Composition features compressed to 8 components

Surrogate

Gaussian process on log PF

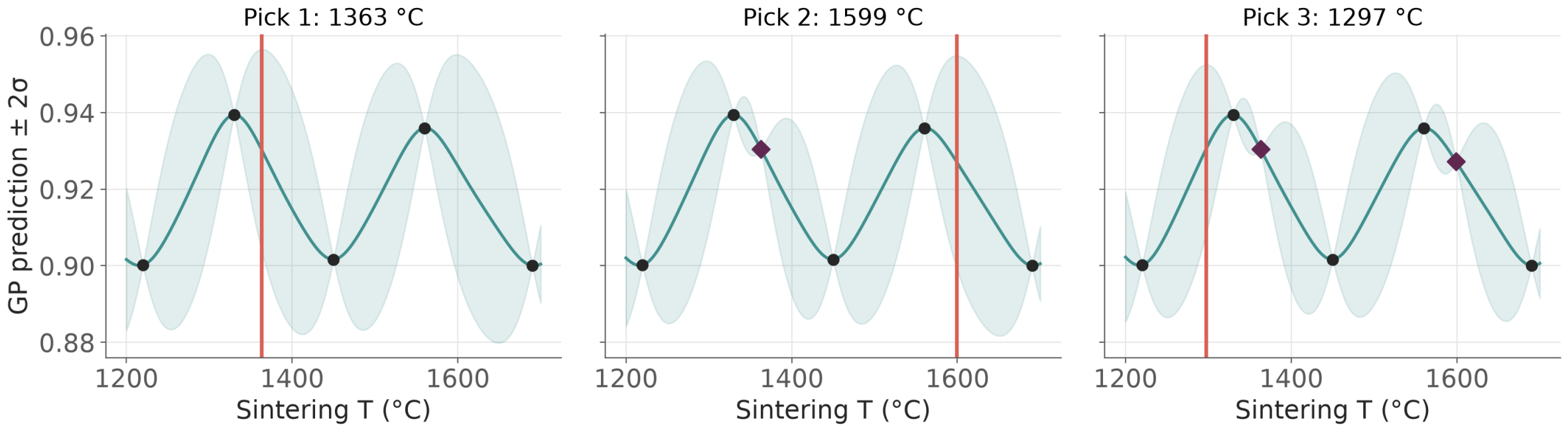
Batches

5 compositions per round, 10 rounds

Same data as the Session 3 challenge, now used to decide what to make

Batches: pretend the model is right, then pick again

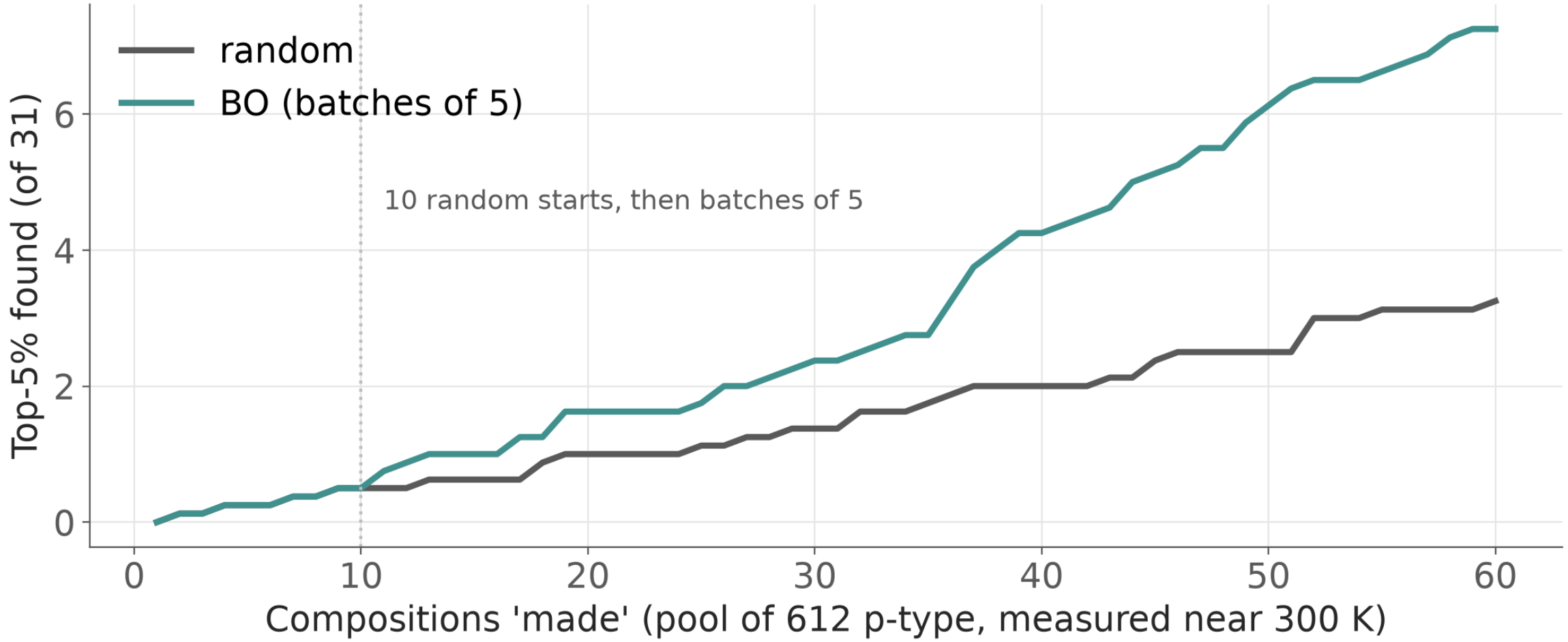
'Kriging believer': after each pick, add the GP's prediction as fake data so the next pick goes elsewhere



Simulated sintering response made for this course.

BO found 7.2 of 31 top compositions in 60 tries; random found 3.2

8 campaigns each; top 5% by measured room-temperature power factor



Data: sysTEm, Tang et al., Integr. Mater. Manuf. Innov. 15, 181 (2026); HHI: Gaultois et al., Chem. Mater. 2013 (via pymatgen). Analysis made for this course.

Constraints you can compute just filter the list; others need a model

Known before the experiment

Supply risk, cost, toxicity, fractions sum to 100%

→ filter or construct the candidates

Learned by experimenting

Cracks? Densifies? Single phase?

→ model the probability of success, weight EI by it

Never let the optimizer propose something you already know you can't make

Scalarization merges two goals into one score, with weights set up front

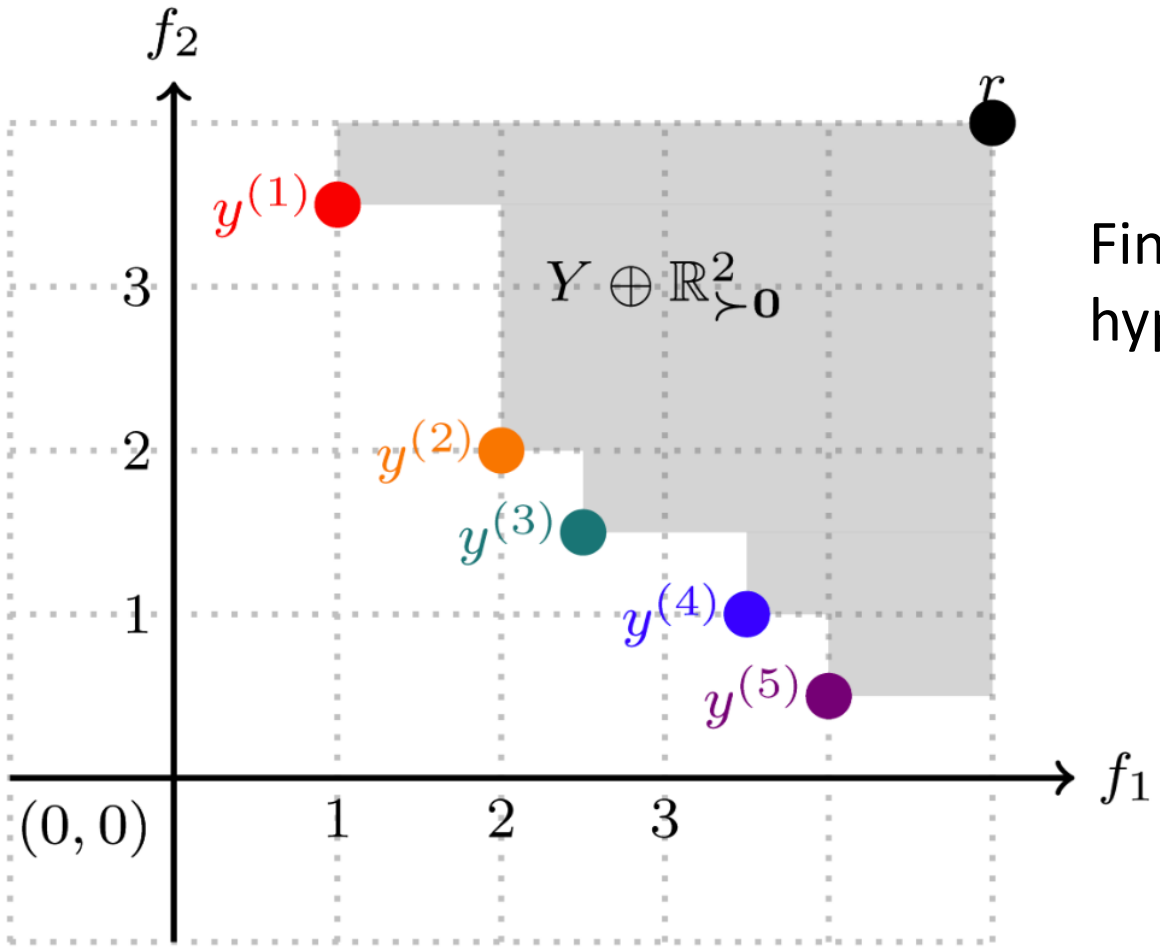
Score = (λ)(Strength Score) + (1-λ)(Cost Score)

Where λ is the relative weight of each score

This can be as simple or as complex as you want.

What is the strength and weakness of this approach?

Hypervolume scores a whole Pareto front so BO can push it outward

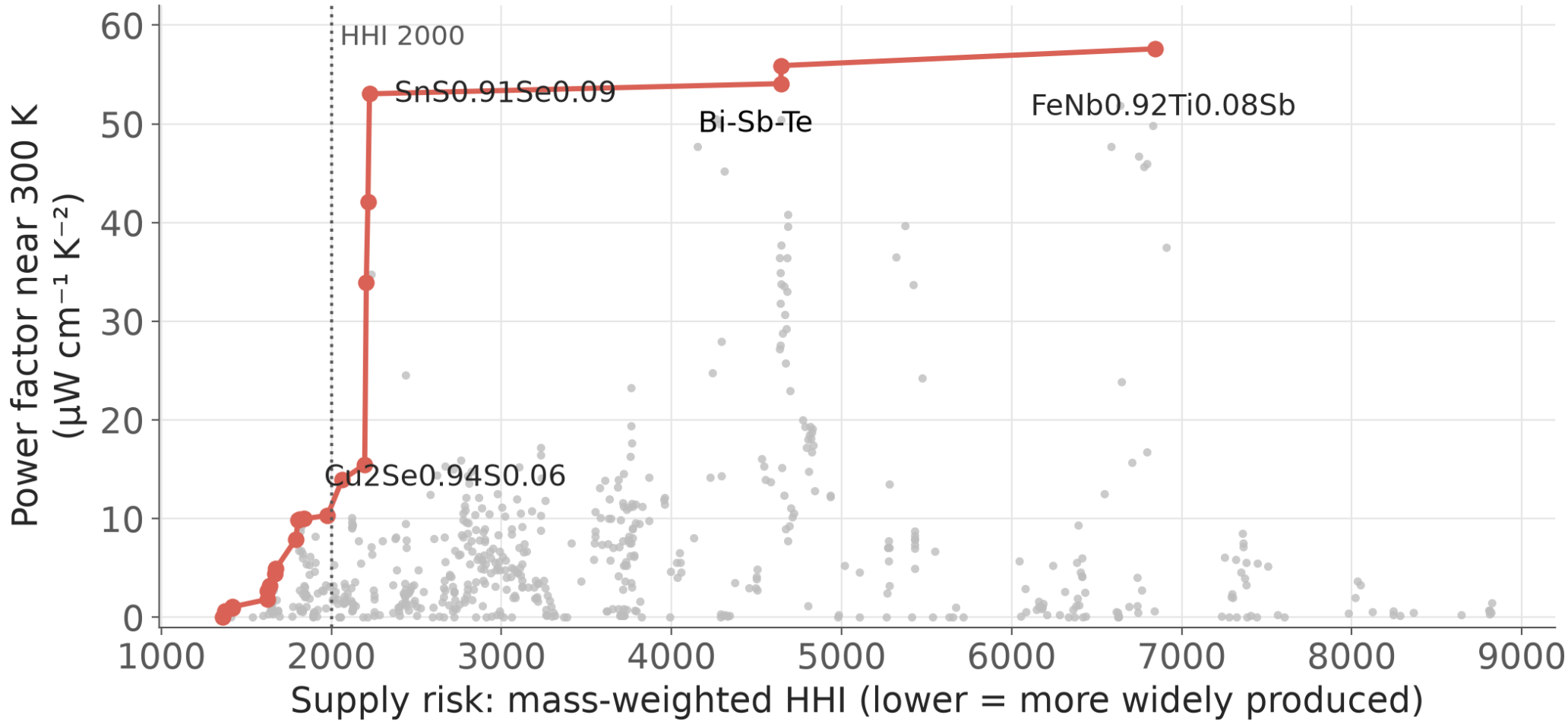


Find points that will maximize the hypervolume.

Figure Referenced From: *https://ax.dev/tutorials/multiobjective_optimization.html*

$SnS_{0.91}Se_{0.09}$: 92% of the best PF at half the supply risk

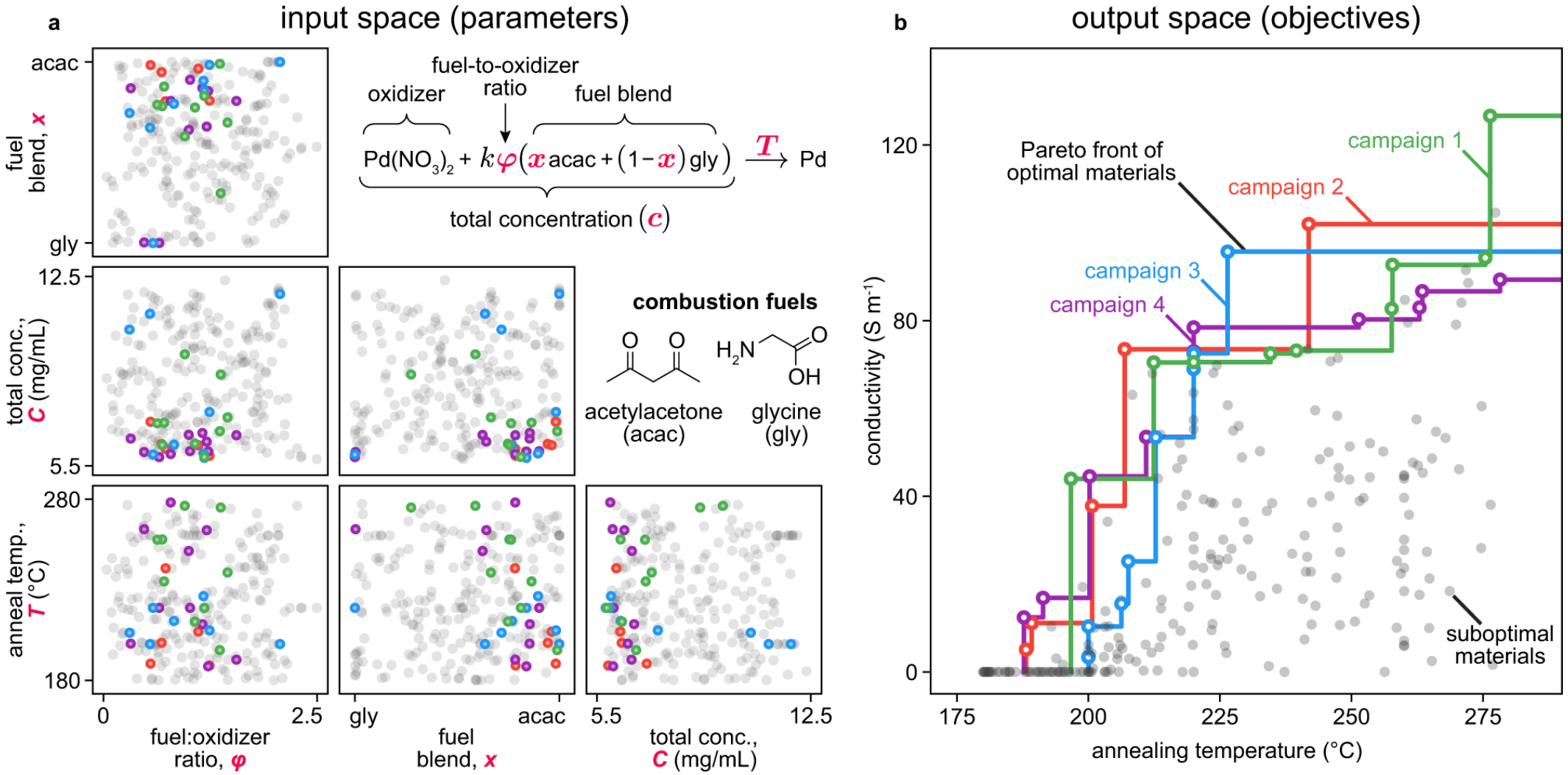
Pareto front: no composition beats these in both power factor and supply risk



Data: sysTEm, Tang et al., Integr. Mater. Manuf. Innov. 15, 181 (2026); HHI: Gaultois et al., Chem. Mater. 2013 (via pymatgen). Analysis made for this course.

With two goals, BO maps a Pareto front, not a single winner

Spray-coated follow-up: 1.1e5 S/m at 191 C and 2.0e6 S/m at 226 C, on par with sputtered Pd



MacLeod et al., Nat. Commun. 13, 995 (2022) https://doi.org/10.1038/s41467-022-28580-6

Optimizing formulations

When the inputs have to add up to 100%

In a formulation, raising one component lowers the others

The constraint

P3HT + D1 + D2 + D6 + D8 = 100%

like a glaze, a coating or an admixture

Build candidates that obey it

P3HT within 15-97%, the rest split at random among nanotubes

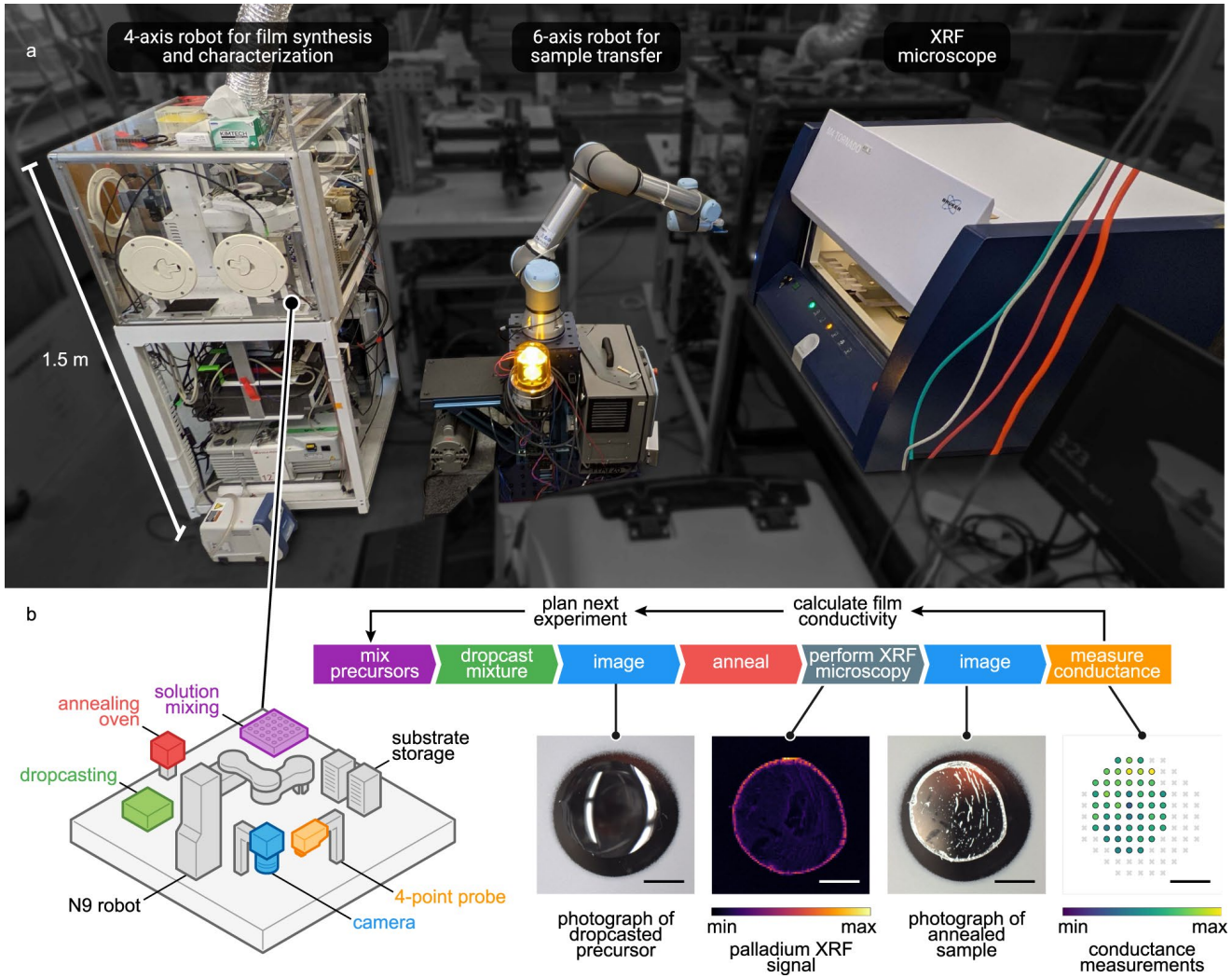
Then BO as usual

Score candidates with EI, pick batches of 4

Ax, BayBE and Honegumi take the same constraint as $x_1 + x_2 + \ldots = 1$

A self-driving lab closes the make-measure-decide loop

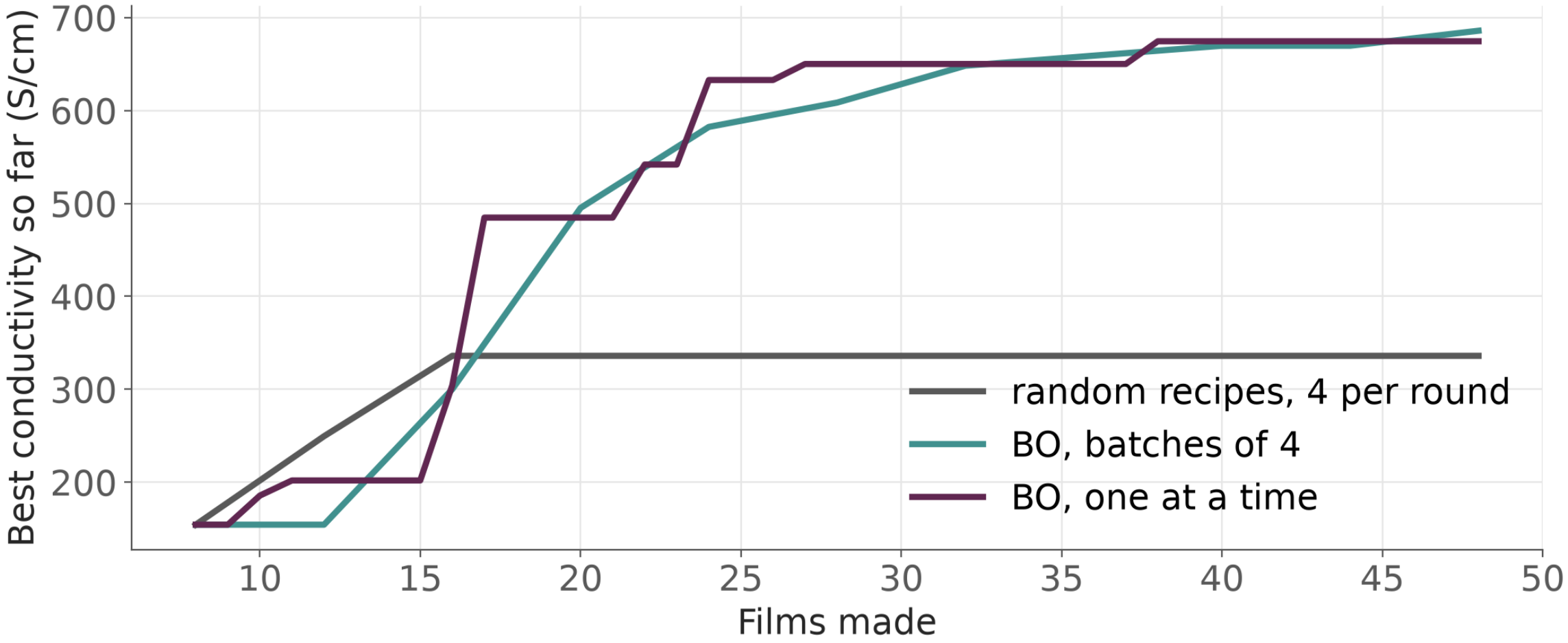
Ada ran 253 combustion-synthesis experiments at about two films per hour, 40-60 unattended



MacLeod et al., Nat. Commun. 13, 995 (2022) https://doi.org/10.1038/s41467-022-28580-6

Batches of 4 matched one-at-a-time BO in a quarter of the rounds

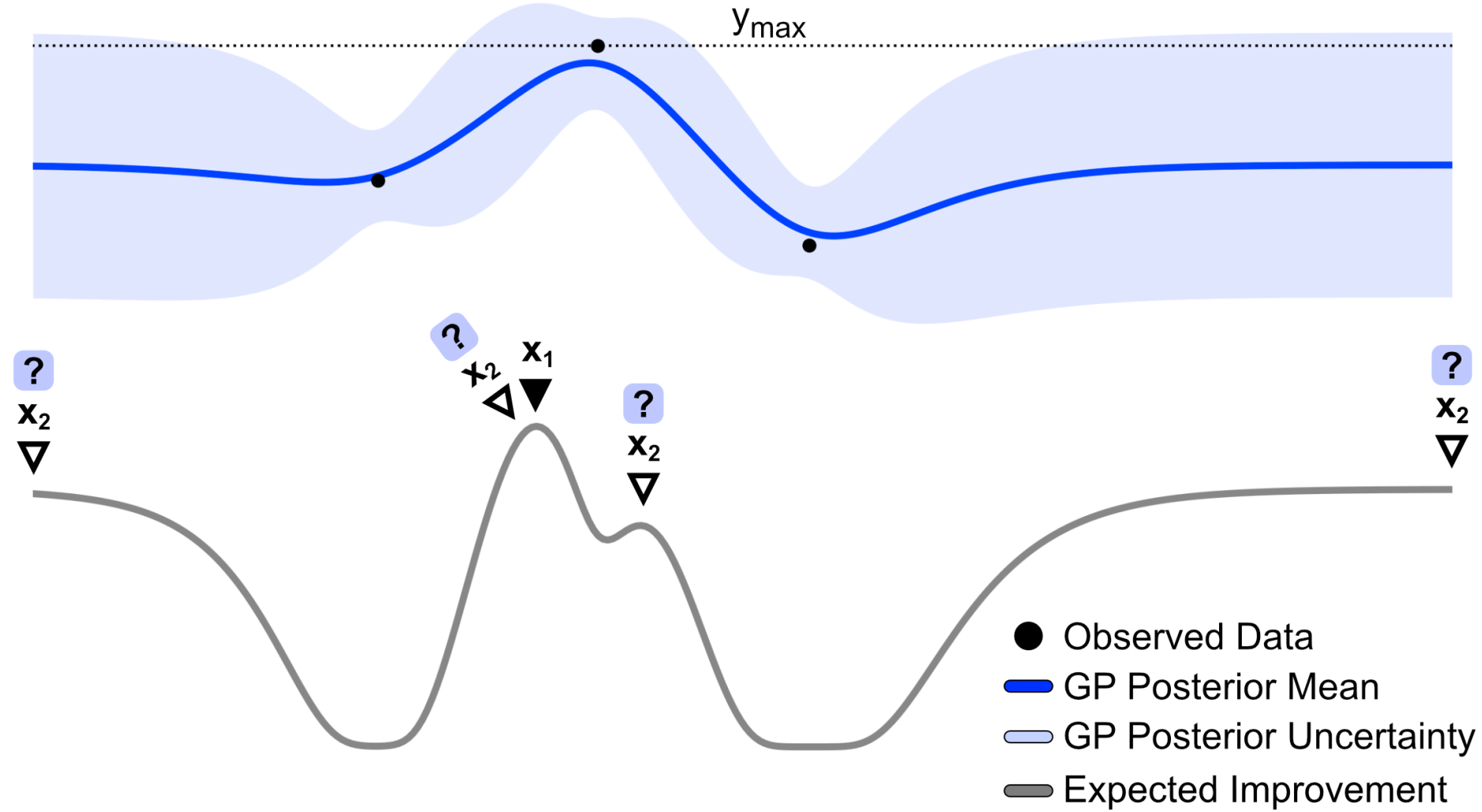
48 films each; random recipes plateau at 336 S/cm, BO reaches about 686



Data: Bash et al., Adv. Funct. Mater. 31, 2102606 (2021), via Liang et al. 2021 (MIT). Emulator and analysis made for this course.

Choosing a batch is harder than choosing one experiment

qEI picks all q points jointly; Kriging believer and constant liar are cheaper heuristics



Honegumi docs, github.com/sgbaird/honegumi (MIT License) https://github.com/sgbaird/honegumi/blob/main/docs/curriculum/concepts/batch/single-vs-batch.md

Describe a formulation by its ingredients, as CBFV describes a compound

Coatings: pigment absorption and scattering add up by concentration (Kubelka–Munk, Duncan's rule)

Raw-materials table

One row per ingredient: datasheet values plus one measurement each (BET, density, absorption and scattering spectra)

→

× fractions

Fraction-weighted average of every property, like CBFV's element averages

→

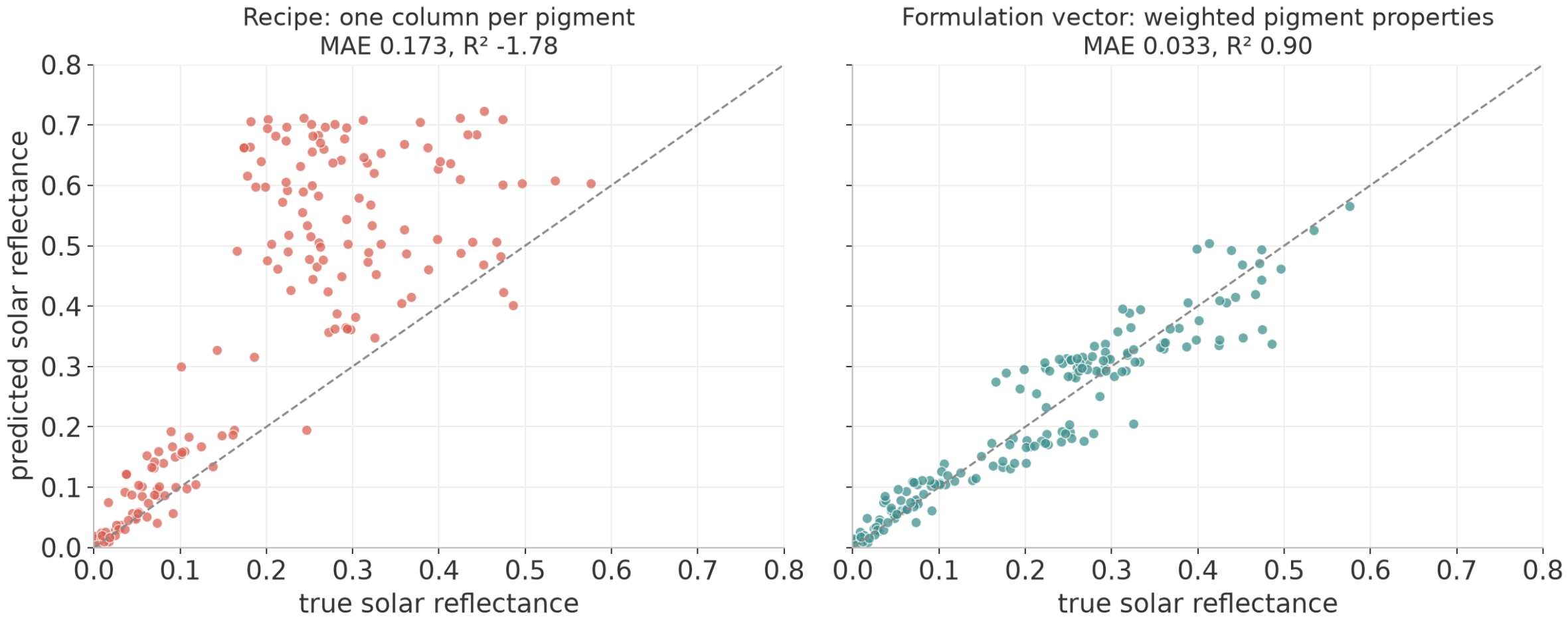
+ processing

PVC, film thickness, firing temperature, dwell time as ordinary columns

The model learns what pigments do, not just which ones you used, so it can rank a pigment you haven't bought yet

A pigment it never saw: recipe columns fail (R^2 -1.8), ingredient properties work (R^2 0.90)

Trained on 232 simulated coatings without an IR-reflective black pigment, tested on the 168 that contain it



Simulated with Kubelka–Munk theory and stylized pigment spectra (not measured data). Made for this course.

Simulations are experiments too: cheap ones

Simulation as the experiment

BO picks which casting or sintering conditions to simulate

Surrogate of a simulation

Fit a fast model to past runs; optimize on it, verify a few

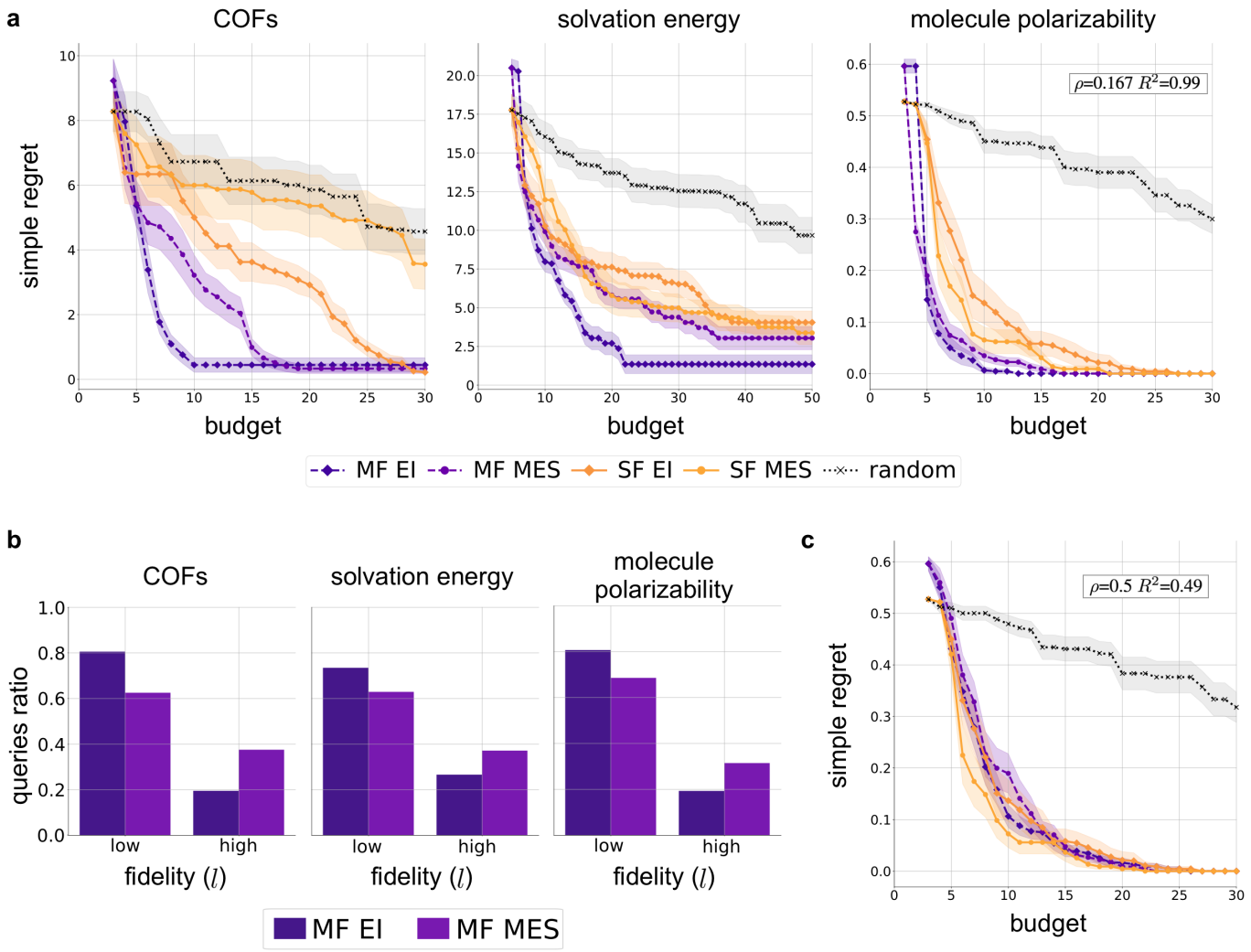
Multi-fidelity

Many cheap runs + a few expensive ones, linked by one model

Spend the expensive runs where the cheap ones can't decide

A cheap proxy helps only if it is cheap and informative

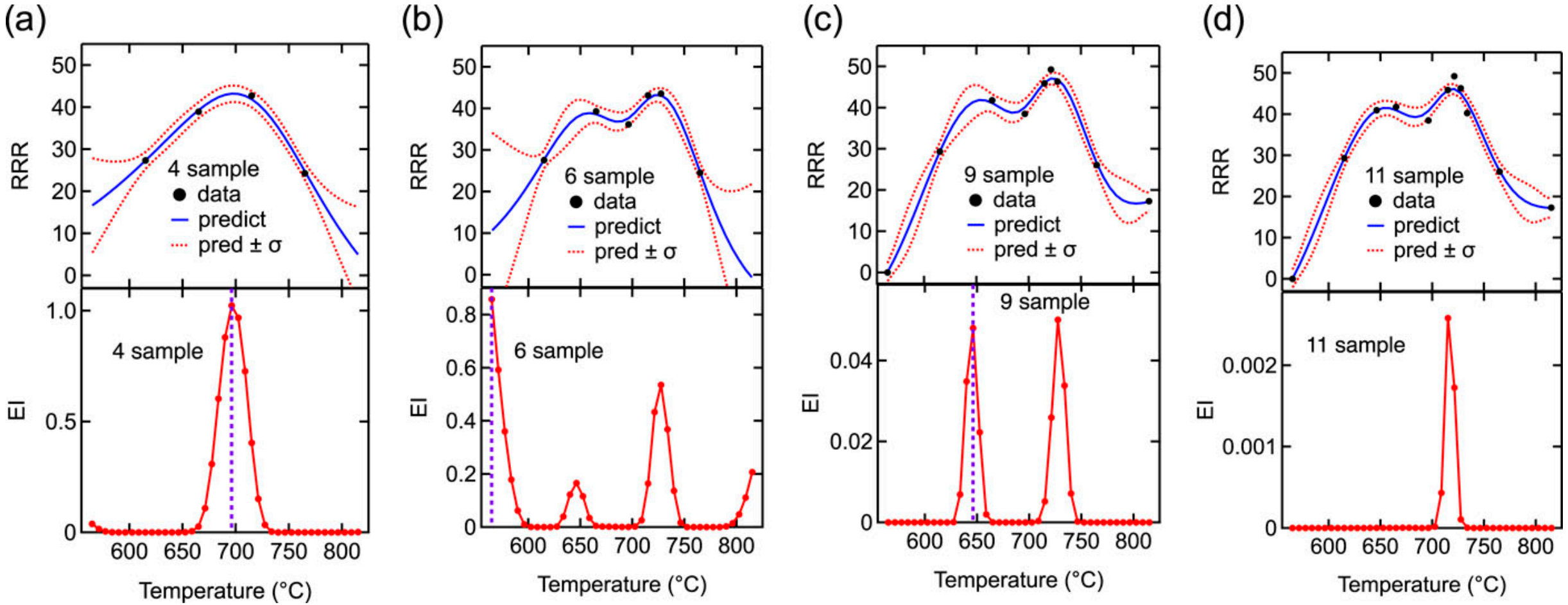
Paper guideline: use multi-fidelity BO when cost ratio < 0.1 and proxy R2 > 0.8



Sabanza-Gil et al., arXiv:2410.00544v3 (2024); Nat. Comput. Sci. (2025) https://doi.org/10.48550/arXiv.2410.00544

BO tuned oxide-film growth conditions in 24 runs

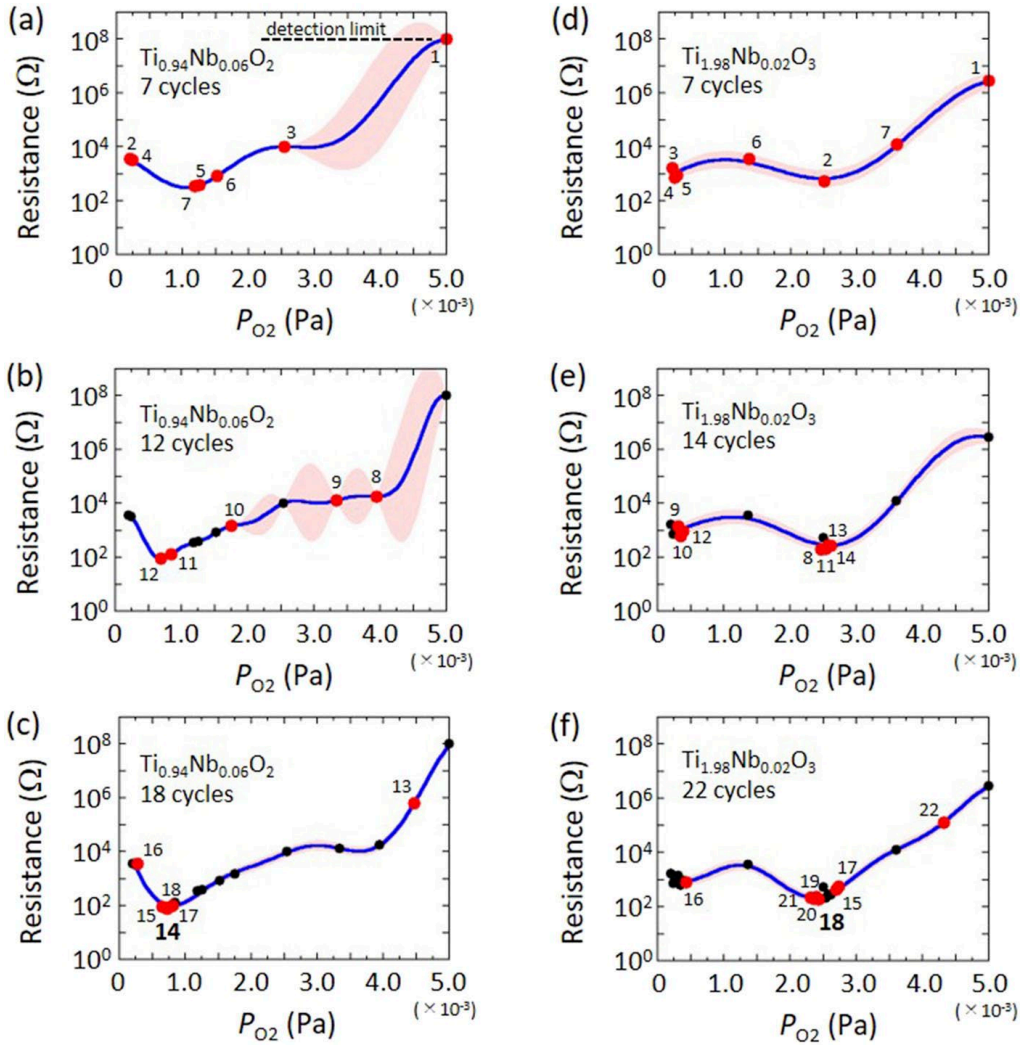
Sequential 1D BO of flux, temperature and O3 nozzle gave RRR above 50 in only 24 MBE growth runs



Wakabayashi et al., APL Mater. 7, 101114 (2019) https://doi.org/10.1063/1.5123019

Robotic sputtering plus BO found the lowest-resistance film

Minimum reached on deposition 14 at about 2 h per film; authors estimate a tenfold throughput gain



Shimizu et al., APL Mater. 8, 111110 (2020) https://doi.org/10.1063/5.0020370

Running a campaign in your lab

Tools and habits

Six habits for a real optimization campaign

Start space-filling

5-10 Latin hypercube or Sobol runs, not one factor at a time

Batch like your lab

One furnace load or coating session per round

Encode constraints

Never accept an impossible suggestion

Human in the loop

Veto unsafe or silly picks; write down why

Log every run

Failures are data for the model

Decide when to stop

Budget, target value or diminishing returns

BO is a decision aid, not an autopilot

You don't have to write BO yourself

Honegumi
Tick boxes for your problem, get a working Ax script

Ax / BoTorch
Batches, constraints, multi-objective, multi-fidelity

BayBE
Lab campaigns with mixtures and substances

scikit-optimize
Simplest pure-Python BO

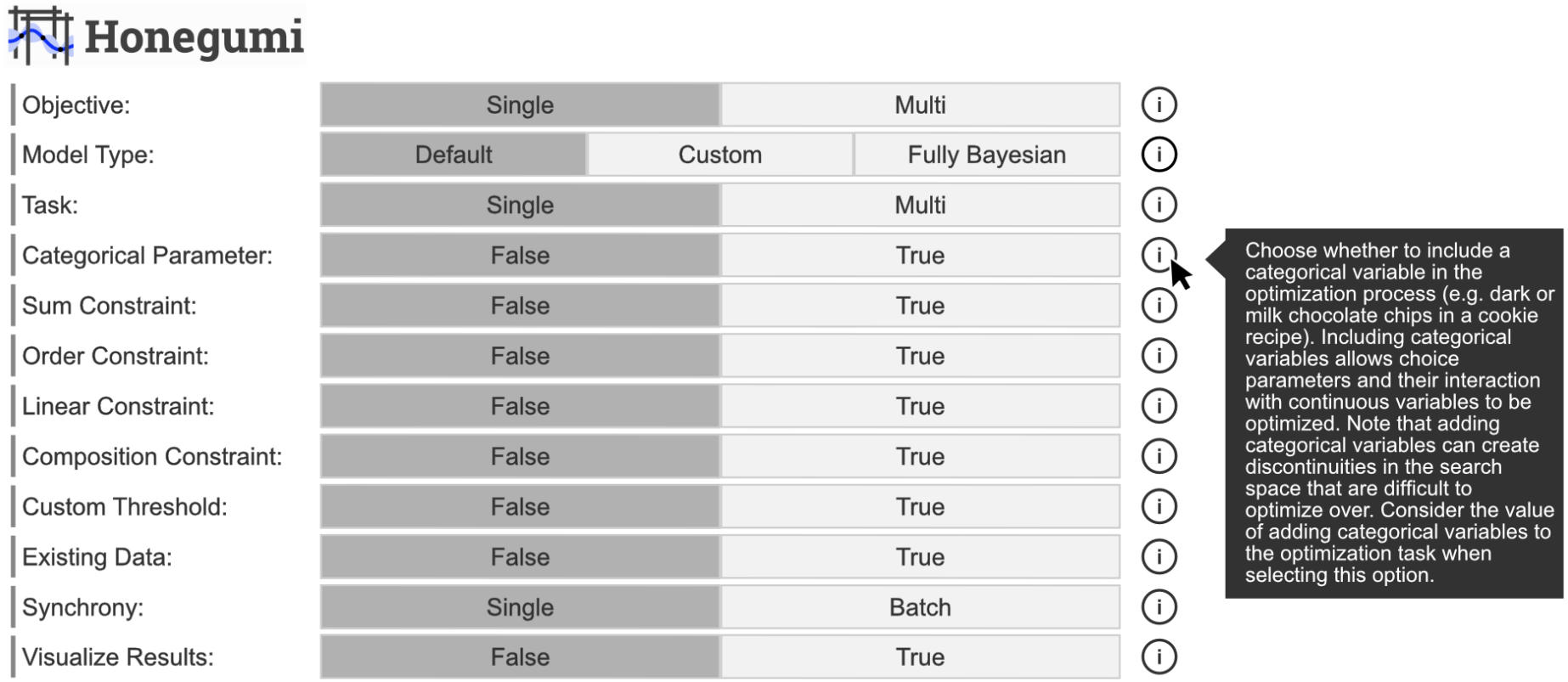
Today's notebook
GP + EI in 20 lines, to see the moving parts

Your process
2-5 inputs, ranges, one objective: start there

honegumi.readthedocs.io

Honegumi turns BO setup into a menu of toggles

Each toggle combination renders a runnable Ax script; all 4,096 combinations are unit tested



Baird, Falkowski & Sparks, arXiv:2502.06815 (2025) https://doi.org/10.48550/arXiv.2502.06815

Four things to remember from today

1. Fewer experiments, not perfect predictions

The model only has to rank where to look

2. Explore on purpose

Greedy search gets stuck on the first peak

3. Match the loop to your lab

Batches, constraints, several goals

4. Trade-offs are decisions

The Pareto front shows the options; you choose

Next session: LLMs, agents and the end-to-end capstone

Hands-on: run your own campaigns

45 minutes · Google Colab · no installation needed

Open the Session 5 notebook in Google Colab

1

Go to the course repo

github.com/sp8rks/ACerS_AI_ML_Workshop

2

Click 'Open in Colab'

Session 5: 05_bayesian_optimization

3

File → Save a copy in Drive

Keeps your edits

4

Runtime → Run all

About 3 minutes; then jump to the part you care about

Exercises: pick the one closest to your work

Process	Materials	Formulation
Escape the local peak: which ξ works?	Maximize PF ÷ supply risk instead of PF	Require ≥ 5% of every nanotube
Smaller budget: is BO still worth it?	Are its picks on the Pareto front?	What does the constraint cost?

New: a pigment the model never saw. Then take a Honegumi script home for your process